

Computer Network (CT 702)

BCT IV/I

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3. Data Link Layer (8 hours)

3.1 Data link layer functions - Framing, error control, flow control, access control

3.2 Error control codes: Parity, checksum, CRC

3.3 Data link protocols: PPP, HDLC

3.4 Logical link control (LLC) and media access control (MAC) sublayers

3.5 Random access protocols: ALOHA, CSMA, CSMA/CD, CSMA/CA 3.6 Token based protocols: Token bus, token ring, FDDI

3.7 Ethernet (IEEE 802.3) and its evolution, bridged and switched Ethernet

3.8 WLAN (Wi-Fi) and IEEE 802.11

3.9 VLAN and its importance

3.10 Virtual circuit switching: ATM, MPLS

Data link Protocol

- Key functions of the Data Link layer (Two sublayer)
 - managing access to whatever medium is in use Media Access Control (MAC),
 - Creating temporary point to point links between a pair of MAC layer addresses to enable data transfer Logical Link Control (LLC)
- Media Access Control (MAC): It deals with actual control of media
 - Construct header and trailer, assembles frames, address and error check, access the medium
- Logical Link Control (LLC): It deals with protocols, flow-control, and error control
 - Interface to upper layer, flow control and error control, management functions
- Point-to-point data transfer
 - Shipping data from one MAC layer address to another

Data link Protocol...

- WAN encapsulation of **frames** at Data Link layer involves one or more of the following services
 - Addressing, a unique destination is required
 - Bit-level integrity check, checksum calculated
 - Delimitation, information boundary for data link frame
 - Protocol identification (PID)

3.1. Functions of Data Link Layer

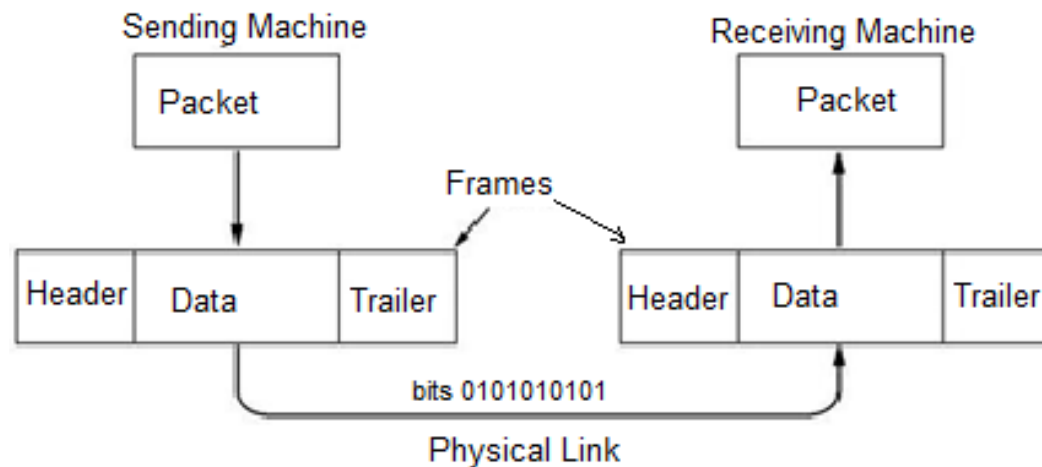
- **Framing**: Data-link layer takes packets from Network Layer and encapsulates them into Frames. Then, it sends each frame bit-by-bit on the hardware. At receiver' end, data link layer picks up signals from hardware and assembles them into frames.
- **Addressing** : Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.
- **Synchronization** : When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.
- **Error Control**: Sometimes signals may have encountered problem in transition and the bits are flipped. These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.
- **Flow Control** : Stations on same link may have different speed or capacity. Data-link layer ensures flow control that enables both machine to exchange data on same speed.
- **Multi-Access** : When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple Systems.

Frame Types and Sizes

- At Data Link layer
 - Protocol data units (PDUs) are called **frames**
- Frame
 - Represents same data that appears in digital form at the Network layer in an IP datagram in the form of whatever sequence of electrical signals map to that data.
- Adding ESC before flag as well as ESC that are occurring in data

Framing

- *Data* - The packet from the Network layer
- *Header* - Contains control information, such as addressing, and is located at the beginning of the PDU
- *Trailer* - Carries error detection or error correction redundant bits.
- **Flag**: signals the start or end of a frame (*SFD & EFD*)



Types of framing

- Fixed size Framing
 - Have Fixed Length
 - No need to define boundaries for Frames
 - It takes **huge memory**
 - Access becomes fast
 - Example
 - ATM Frames (54 byte cells)
- Variable size Framing
 - Two types: Length and End Frame delimiter (EFD)
 - EFD: Character stuffing and bit stuffing
 - Not Fixed Size
 - It takes least memory
 - Access becomes slow
 - Need a way to define the **end of the frame** and the **beginning of the next frame**

What is Error?

- Corruption of data during transmission
 - Bits lost
 - Bits changed
 - Bits added
- Types of errors;
 - Single bit errors
 - Multiple bit errors/ Burst errors

Error Detection (channel coding in DCS)

- Error Detection techniques allow the destination to detect errors.
- Sometimes undetected errors will still remain but the goal is to minimize these errors.
- To detect and correct errors, enough **redundancy bits** need to be sent with data.
- **Redundancy bits** are the extra bits sent by source to inform destination about the data sent.
- Types
 - Parity Check (even or odd)
 - Checksum
 - Cyclic Redundancy Check (based on binary division)
- The **parity generating** technique is one of the most widely used error detection techniques for the data transmission.

Parity Generating Table

Original Data	Even Parity	Odd Parity
0 0 0 0 0 0 0 0	0	1
0 1 0 1 1 0 1 1	1	0
0 1 0 1 0 1 0 1	0	1
1 1 1 1 1 1 1 1	0	1
1 0 0 0 0 0 0 0	1	0
0 1 0 0 1 0 0 1	1	0

Even Parity generator

3-bit message			Even parity bit generator (P)
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

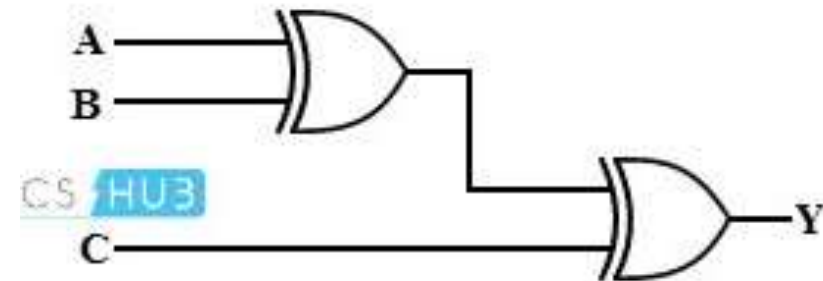
$$P = \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} \bar{C} + A B C$$

$$= \bar{A} (\bar{B} C + B \bar{C}) + A (\bar{B} \bar{C} + B C)$$

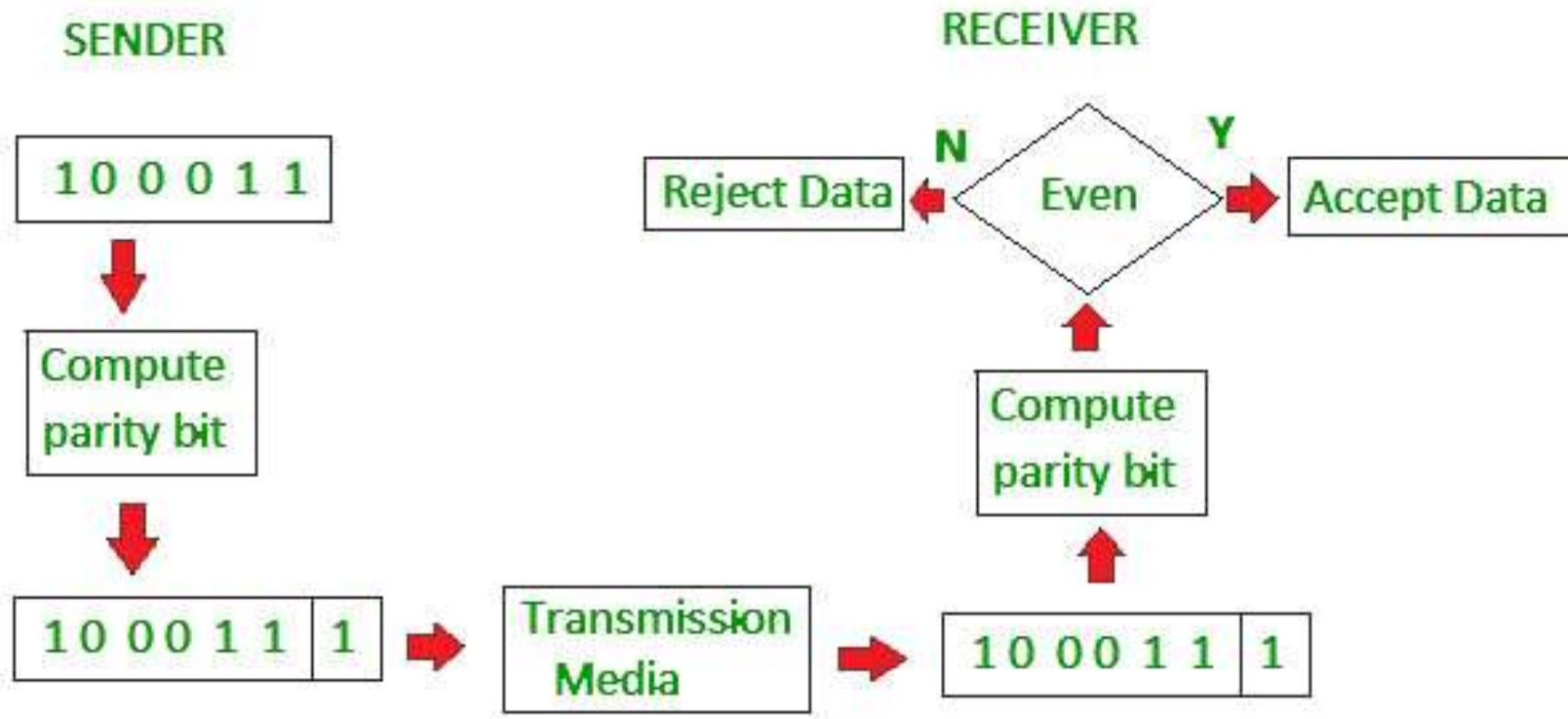
$$= \bar{A} (B \oplus C) + A (\overline{B \oplus C})$$

$$P = A \oplus B \oplus C$$

BC \ A	00	01	11	10
	0	1	3	2
00	0	1	0	1
01	1	0	1	0

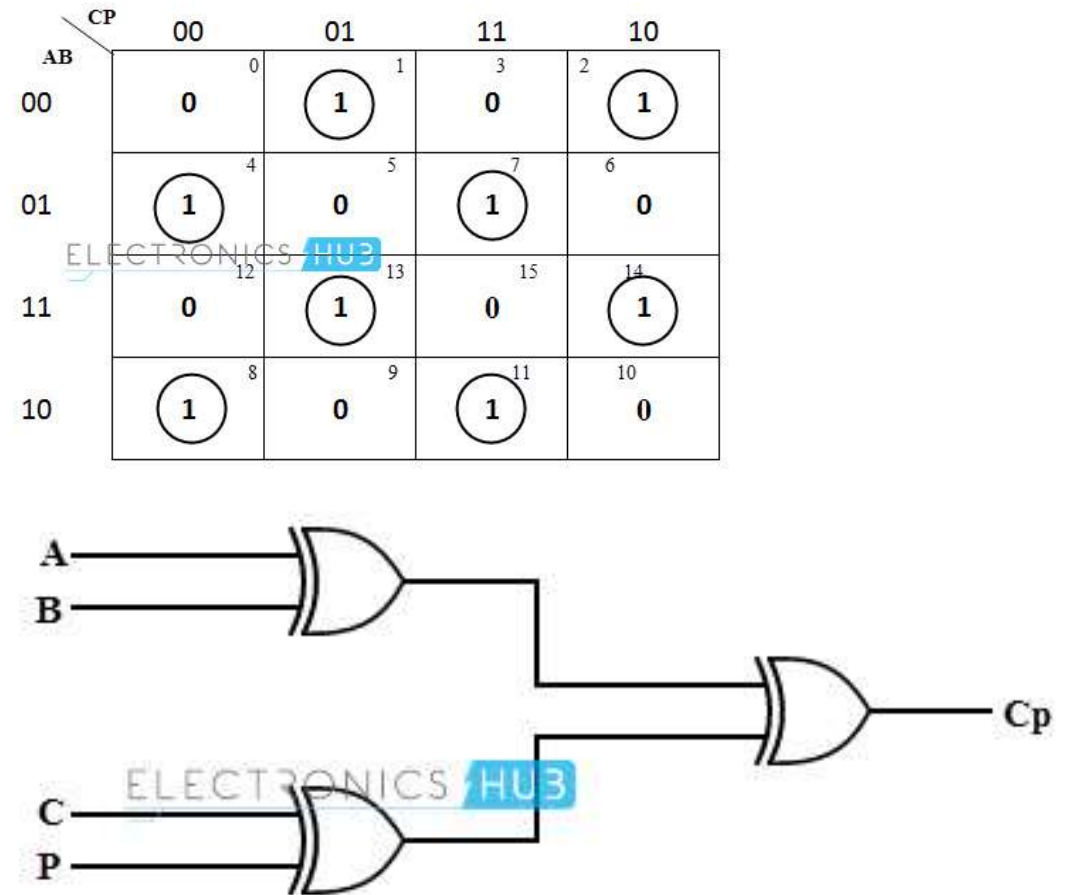


Parity Check: Even parity



Even parity Checker

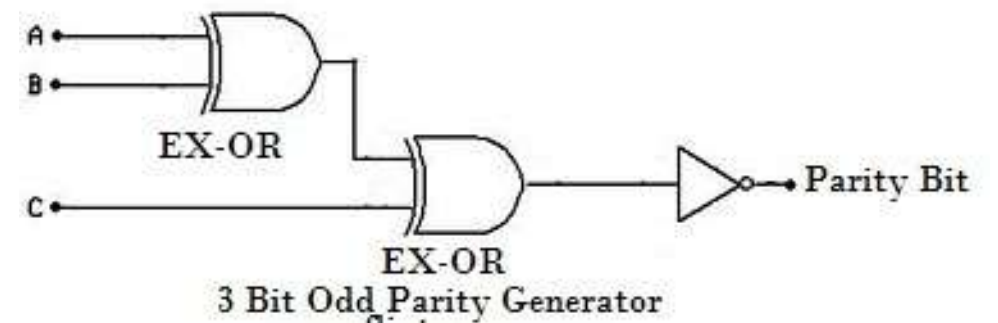
4-bit received message				Parity error check C_p
A	B	C	P	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



Odd parity generator

3-bit message			Odd parity bit generator (P)
A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

A \ BC	00		01		11		10	
	0		1		3		2	
00	1		0		1		0	
01	0		1		0		1	

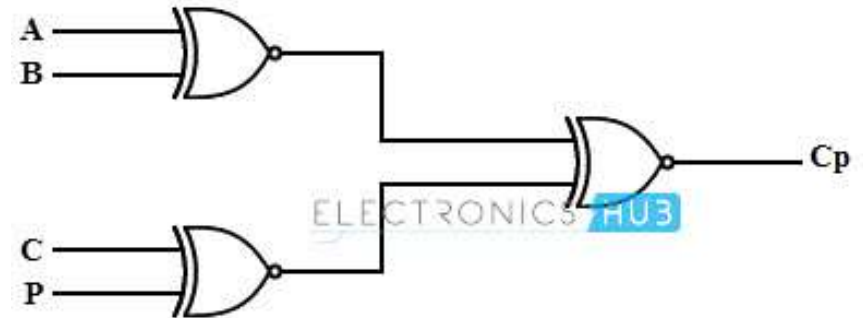


$$P = A \oplus B \text{ Ex-NOR } C$$

Odd parity checker

4-bit received message				Parity error check C_p
A	B	C	P	
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

AB \ CP				
	00	01	11	10
00	0 1	1 0	3 1	2 0
01	4 0	5 1	7 0	6 1
11	12 1	13 0	15 1	14 0
10	8 0	9 1	11 0	10 1



Checksum

- In checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum.
- The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.

Original Data

10011001	11100010	00100100	10000100
----------	----------	----------	----------

1
k=4, m=8

Sender

```

1  10011001
2  11100010
   -----
   101111011
   1
   -----
   01111100
3  00100100
   -----
   10100000
4  10000100
   -----
   100100100
   1
   -----
Sum: 00100101
Checksum: 11011010
    
```

Receiver

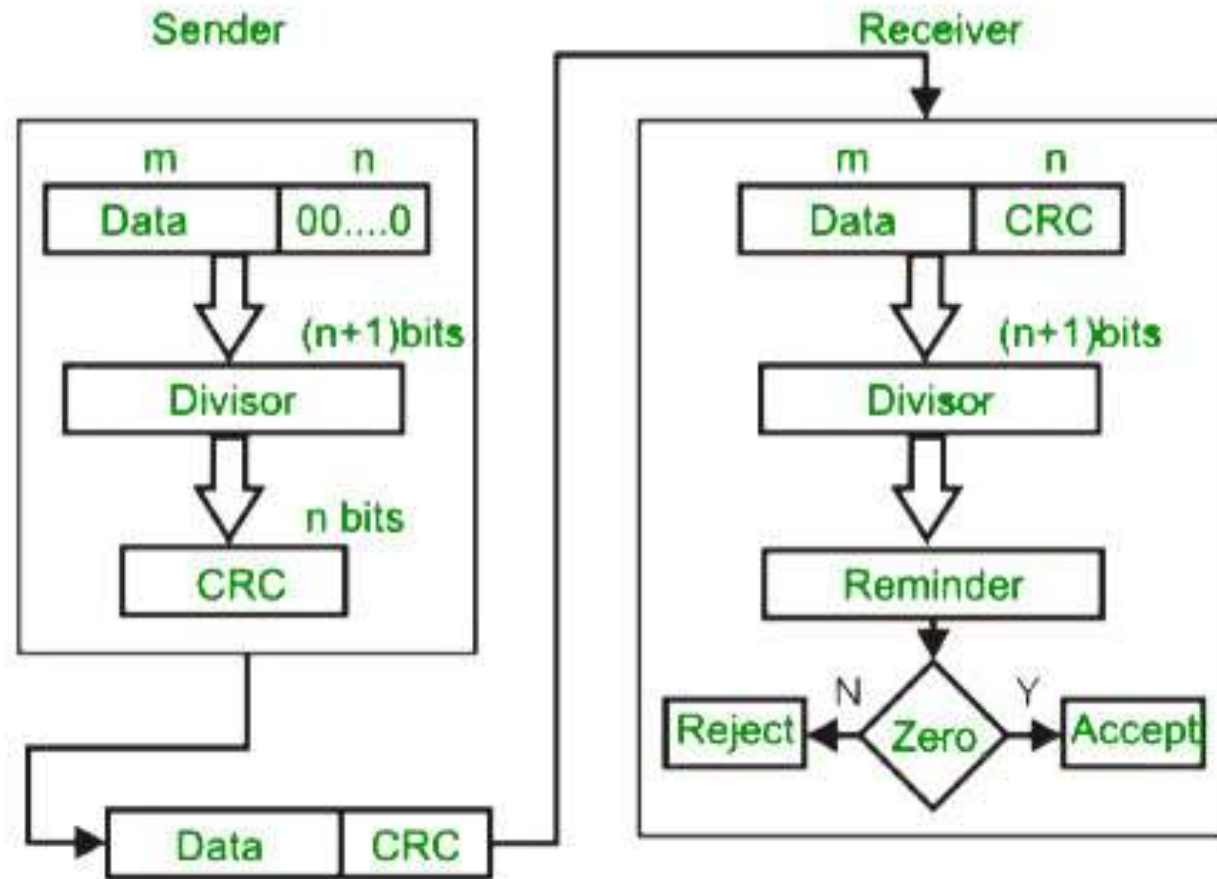
```

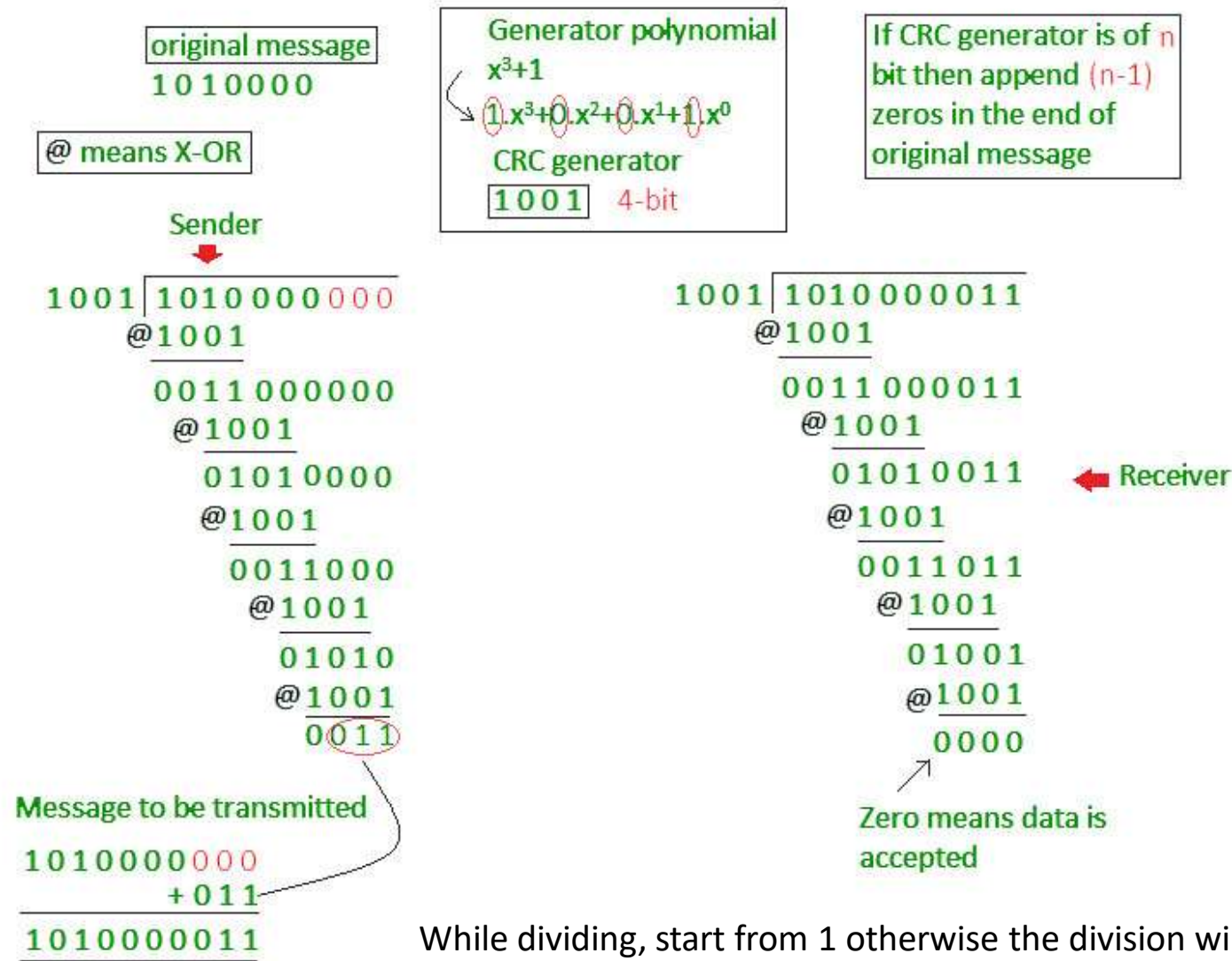
1  10011001
2  11100010
   -----
   101111011
   1
   -----
   01111100
3  00100100
   -----
   10100000
4  10000100
   -----
   100100100
   1
   -----
   00100101
   11011010
   -----
Sum: 11111111
Complement: 00000000
Conclusion: Accept Data
    
```

Cyclic redundancy check (CRC)

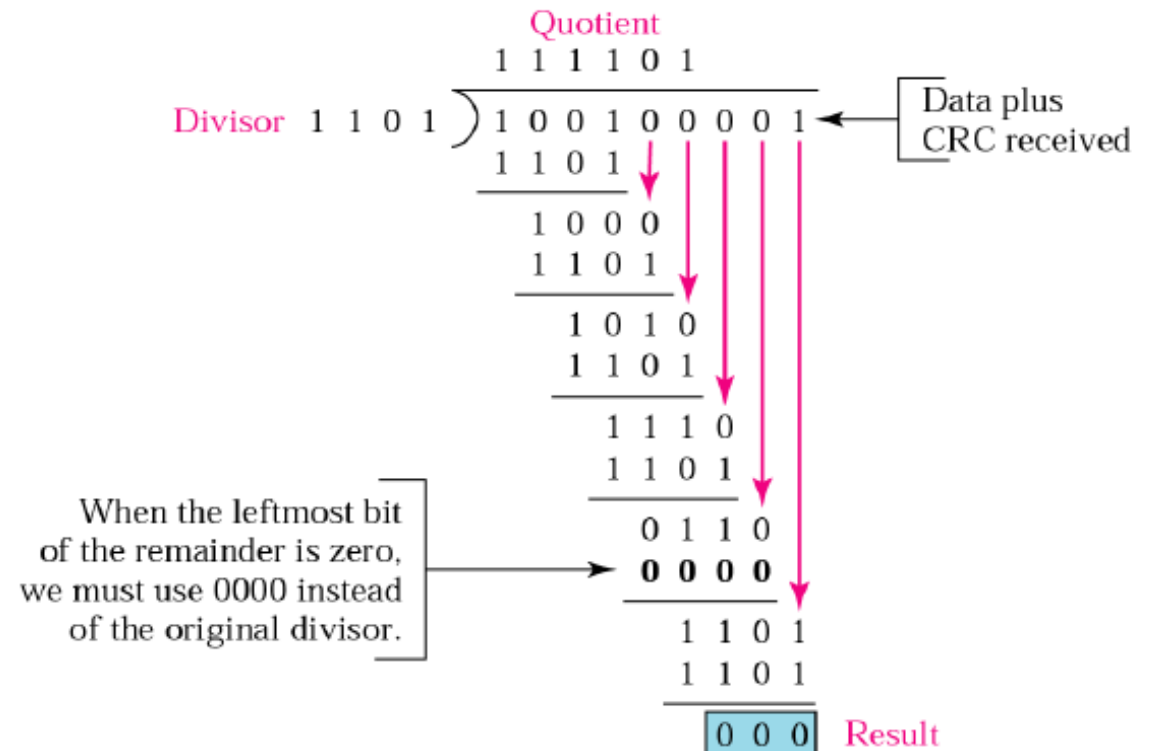
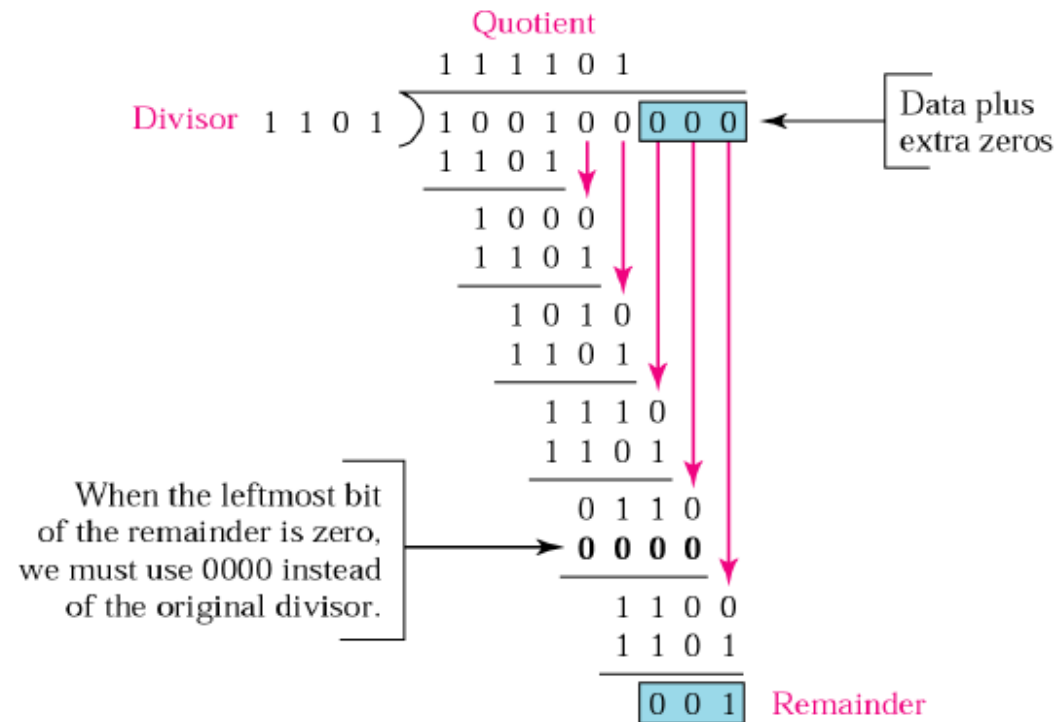
- Unlike checksum scheme, which is based on addition, CRC is based on (Modulo-2) **binary division**, most widely used. CRC bits means redundant bits.
- Append the string of **n** 0s to the data unit where **n** is one less than the number of bits in the predefined **divisor (n+1)** bits long. It is used if direct digital divisor bits (not polynomial) are provided. Also, If divisor is **n**, **n-1** is appended. **If divisor is provided in Polynomial form, number of appending bits=maximum power of polynomial (divisor).**
- Divide the newly generated data unit by the divisor. Quotient is not necessary here.
- The remainder is **n** bit CRC (it should have exactly one less bit than divisor). The last **n** bits in the remainder.
- The CRC bits are **appended** to the **end of data** unit so that the resulting data unit becomes exactly divisible by **predetermined** binary number. This codeword is to be transmitted.
- At the destination, the incoming data unit is **divided** by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

CRC



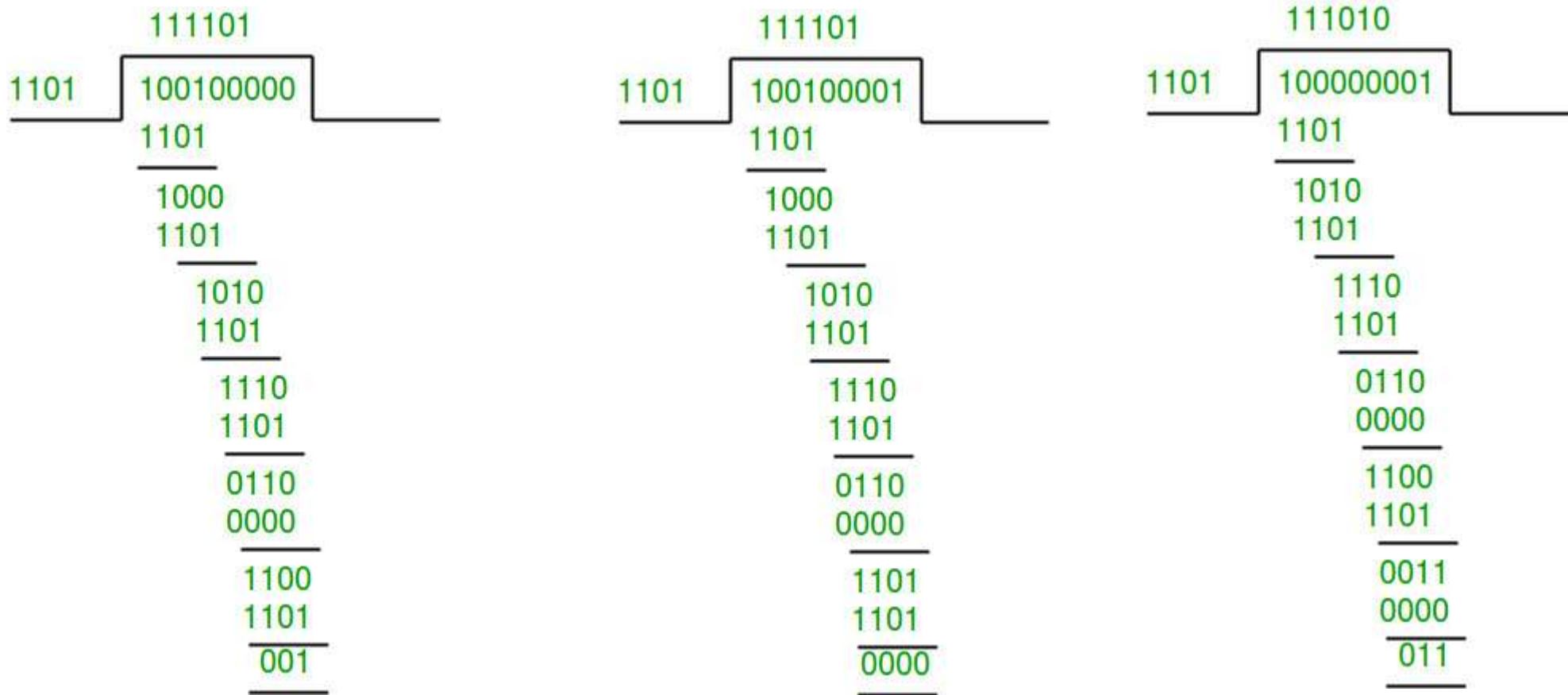


CRC Encoding/Decoding



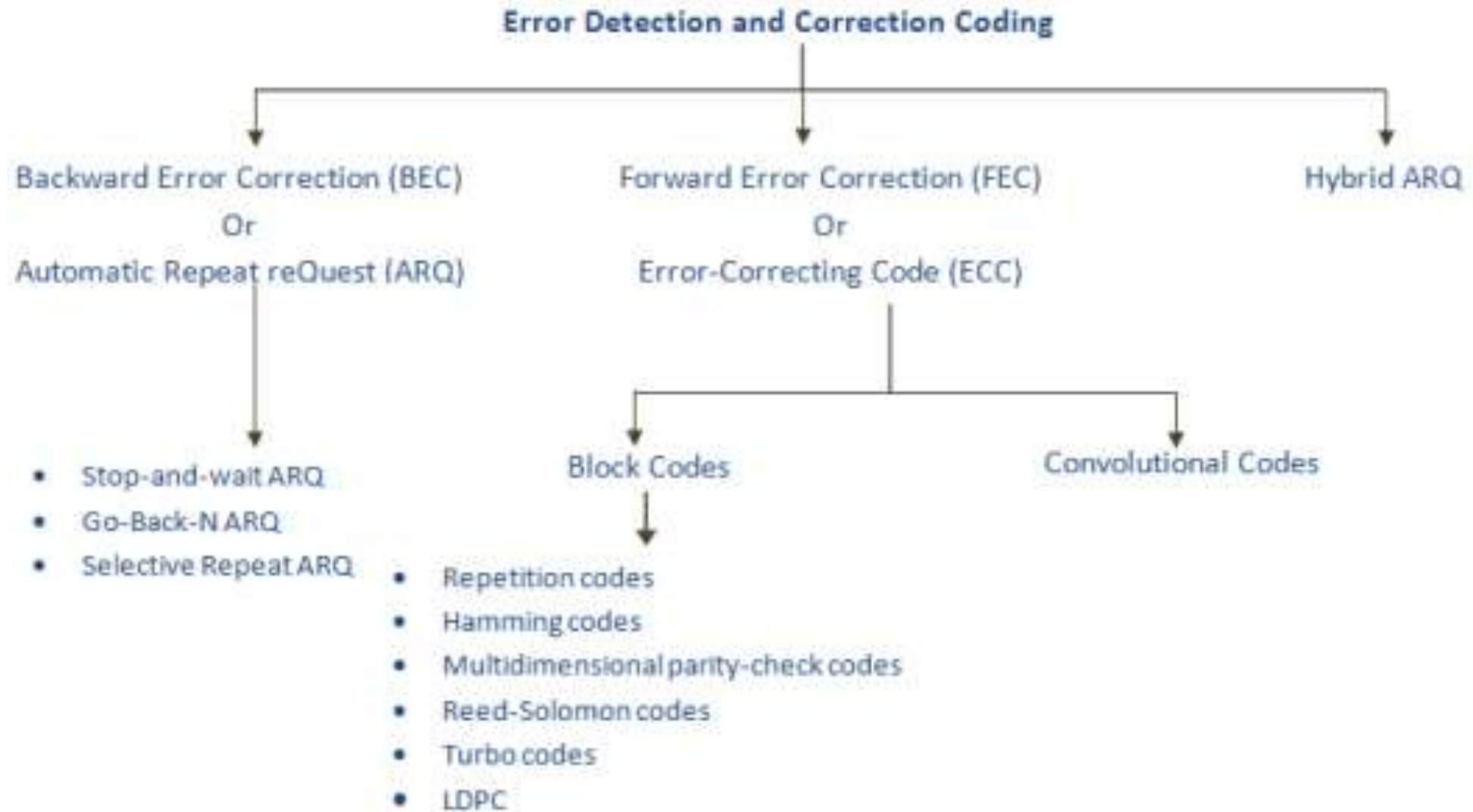
* While dividing, start from 1 otherwise the division will be lengthy.

Example :(No-error/error in transmission)



Practice

- Find CRC of data 10101010 and divisor polynomial $x^4 + x^3 + 1$ or (11001) by binary division (modulo 2) method.
- CRC: 0010
- Codeword: 101010100010 (transmitted)
- Efficiency = data bits / total transmitted bits = $(10/14) * 100\%$
 - channel utilization for sending message



Error control mechanism

- Error correction : This type of error control allows a receiver to reconstruct the original information when it has been added in transmitting site.
- Backward Error Correction or Automatic repeat request (ARR)
- Forward Error Correction
 - Reed-Solomon (RS) block codes
 - Binary Convolutional codes
 - **Hamming codes**

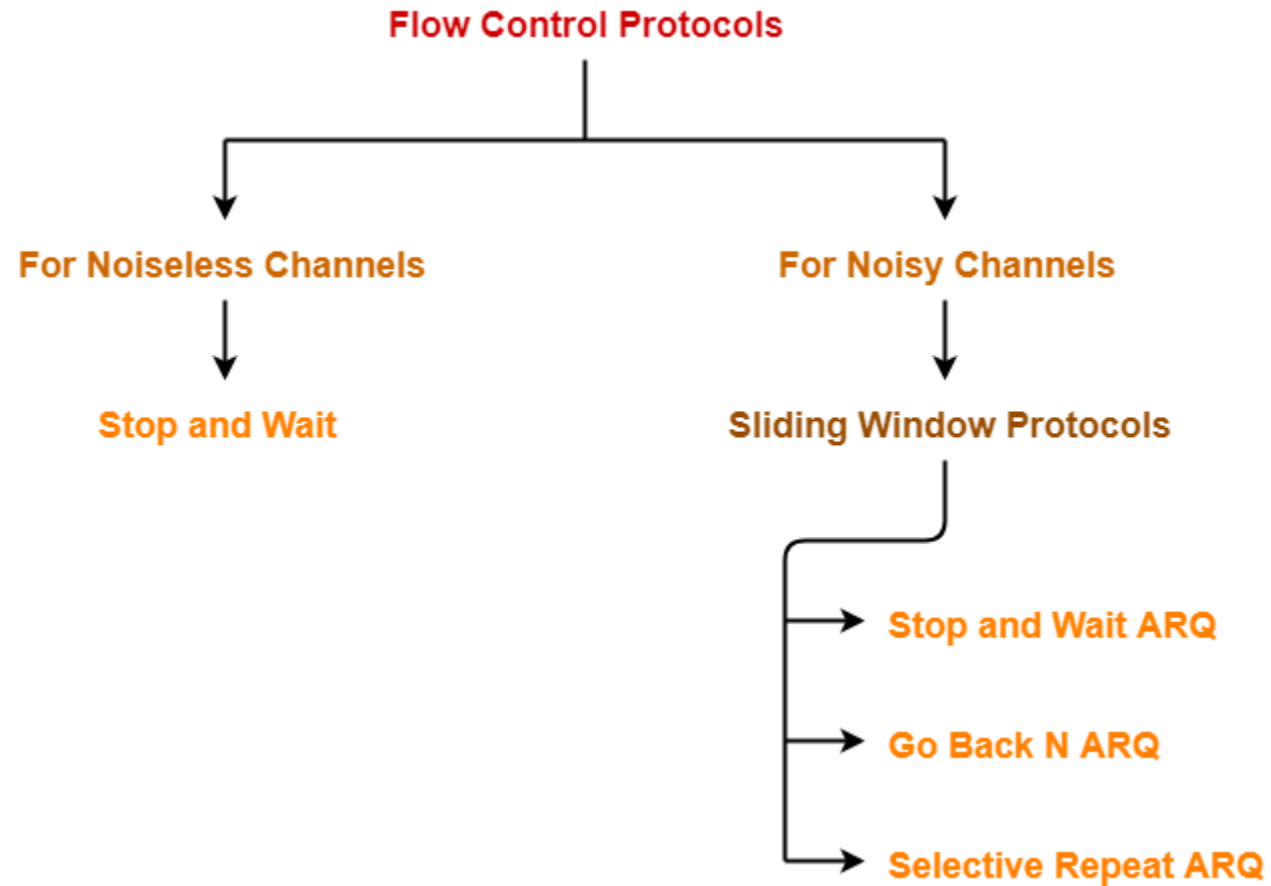
Backward Error Correction

- Known as **Automatic Repeat Request** (ARR)
- Receiver device sends a request to the source device to **re-send** the data after detecting the error or errors
- More often used because it requires less bandwidth
- A return channel is needed for backward error correction
- There are two ways to overcome the errors
 - **Positive acknowledgement:** Receiver returns confirmation of each block received correctly. The transmitter re-sends the block that is not acknowledged.
 - **Negative acknowledgement:** Receiver returns a request **to retransmit only the data with error** (*Selective repeat flow control)

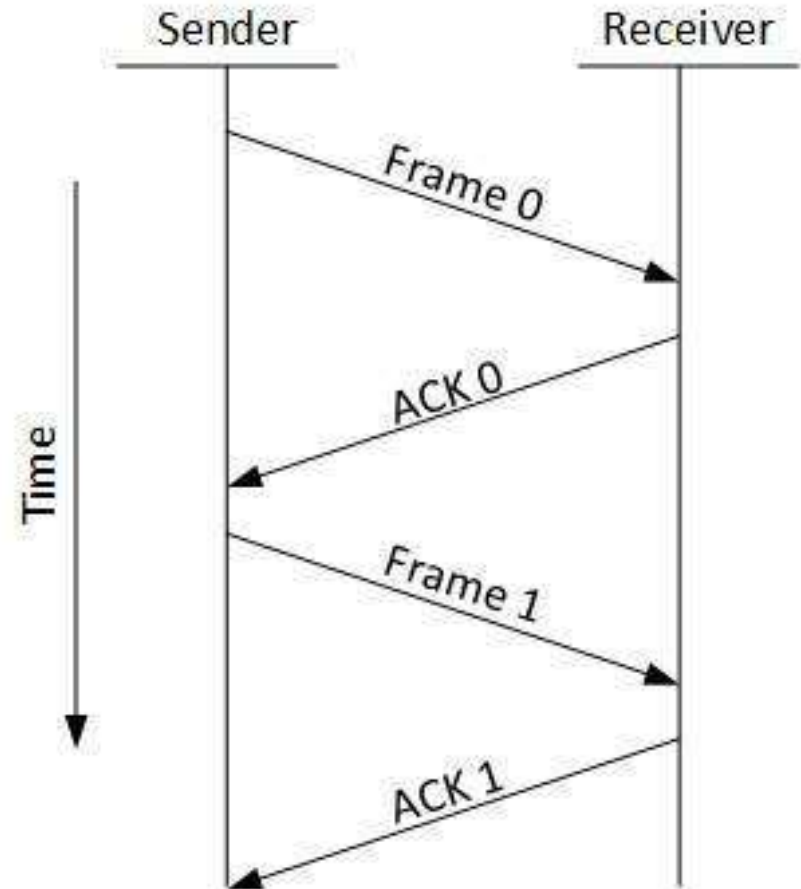
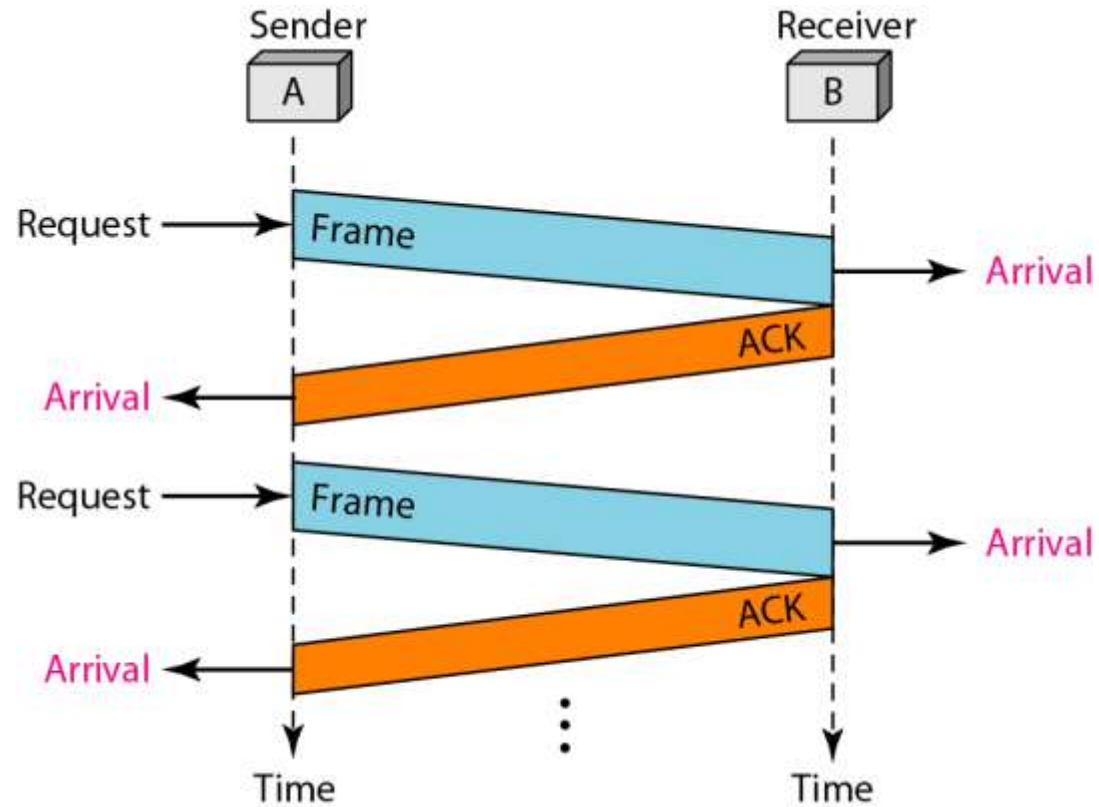
Forward Error Correction

- This technique allows the receiver to **detect and correct** errors without asking the sender for retransmission
- The bandwidth requirement is higher but return channel is not needed
- Redundant data, sent by transmitter is also called **error correction code**
- Redundancy bits are added to the transmitted information using a predetermined information
- Each redundancy bit can be a function of many parts of original data or also can be nonsystematic

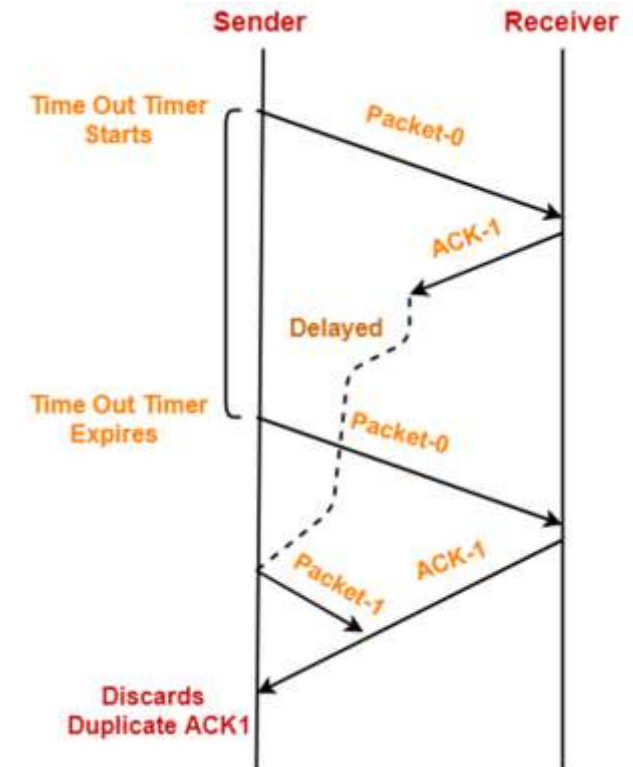
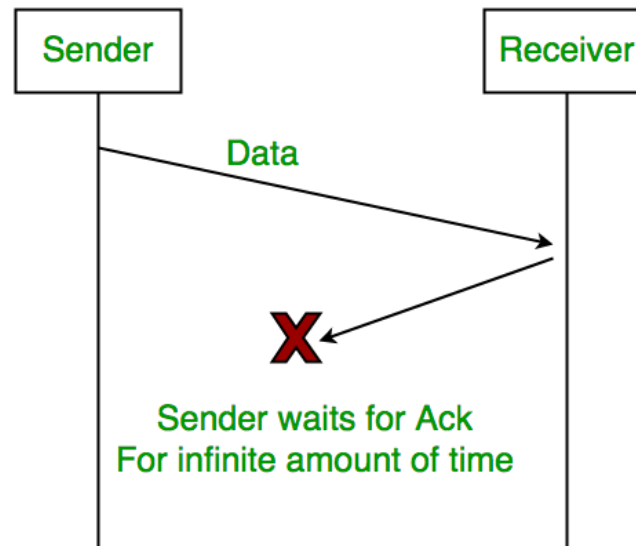
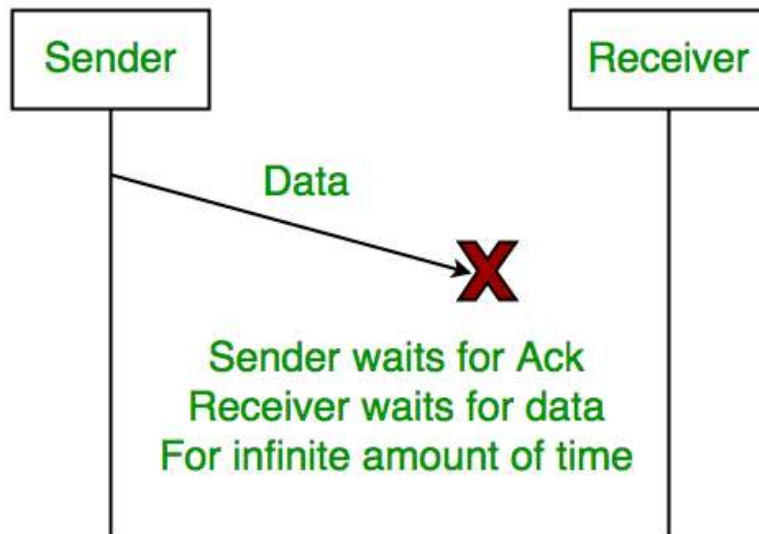
Flow Control ?



Stop & Wait



Problems: Lost data, Lost Acknowledgement & Delayed Acknowledgement

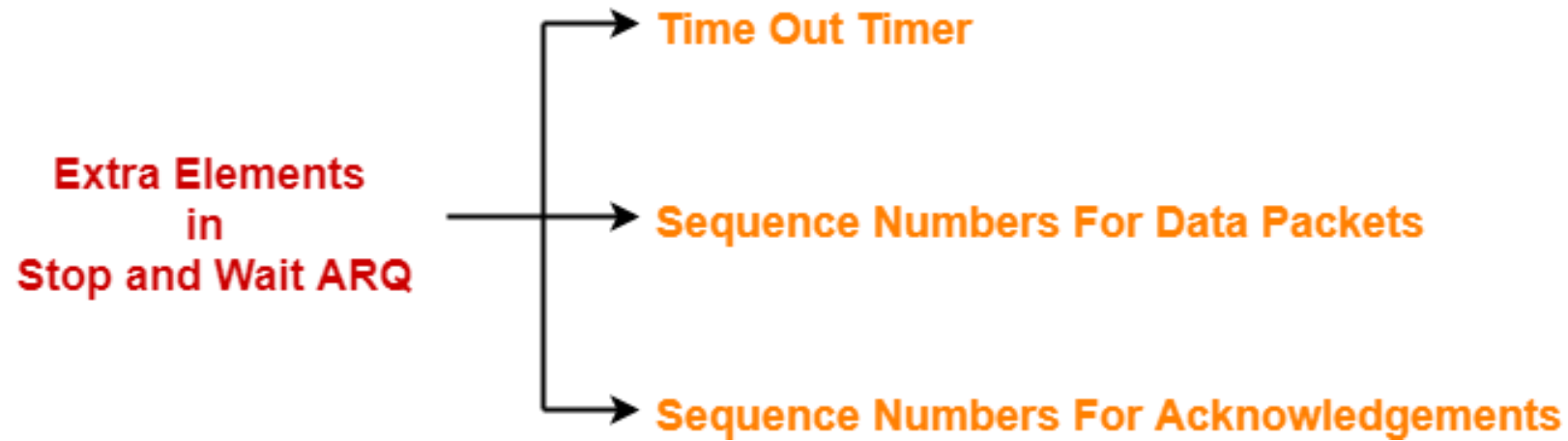


Problems & Solution

Problems/Solution

1. Lost data-Time Out timer
2. Lost Acknowledgement – Sequence No.(on Data)
3. Delayed Acknowledgement:-Sequence No.(on Ack)

Stop and Wait ARQ (Automatic Repeat Request)



Stop (and) Wait + Time Out + Sequence No.(Data) + Sequence No. (ACK)

Lost Data

↑

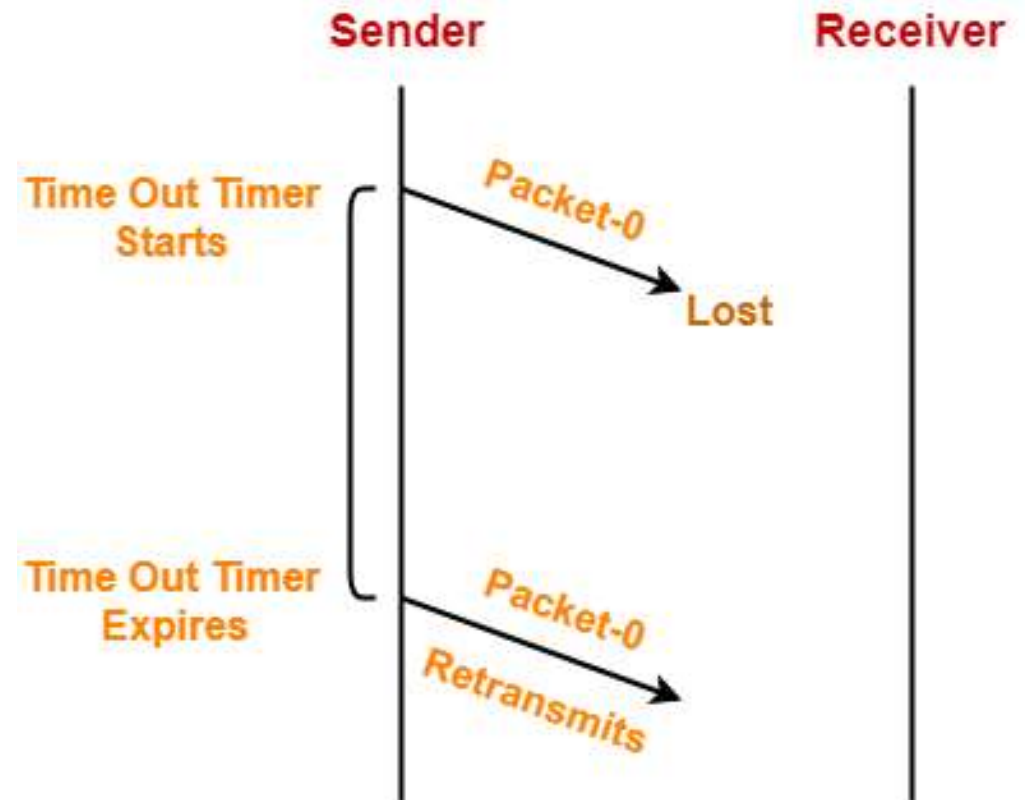
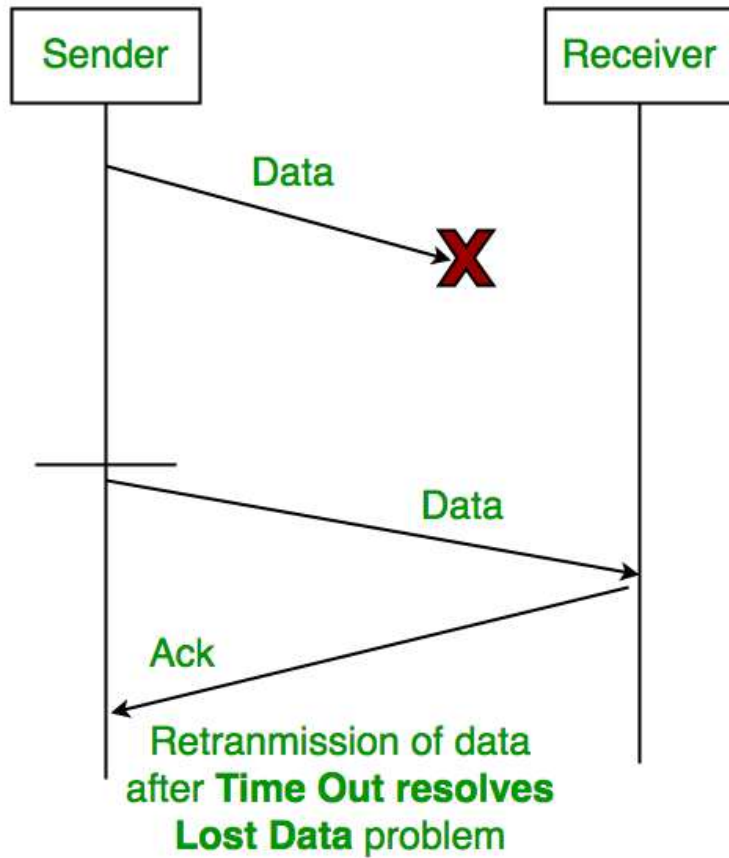
Lost Ack

↑

Delayed Ack

↑

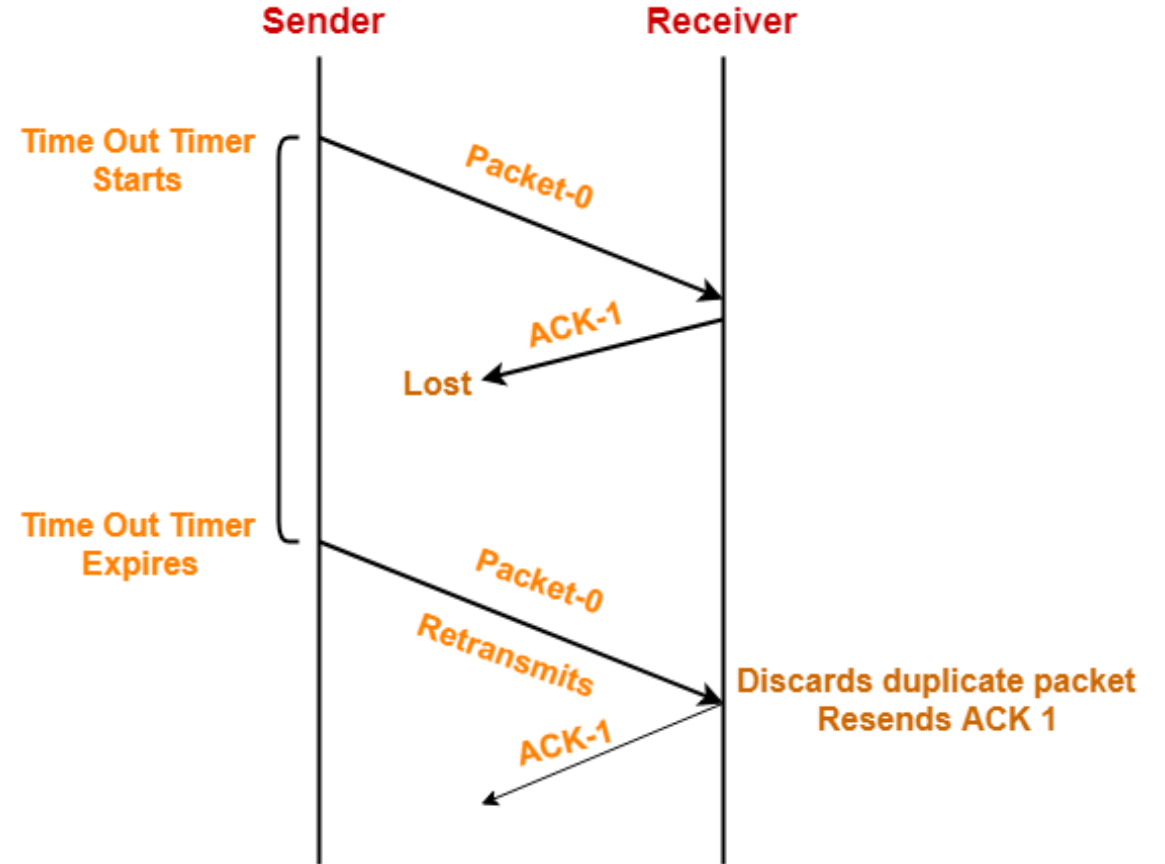
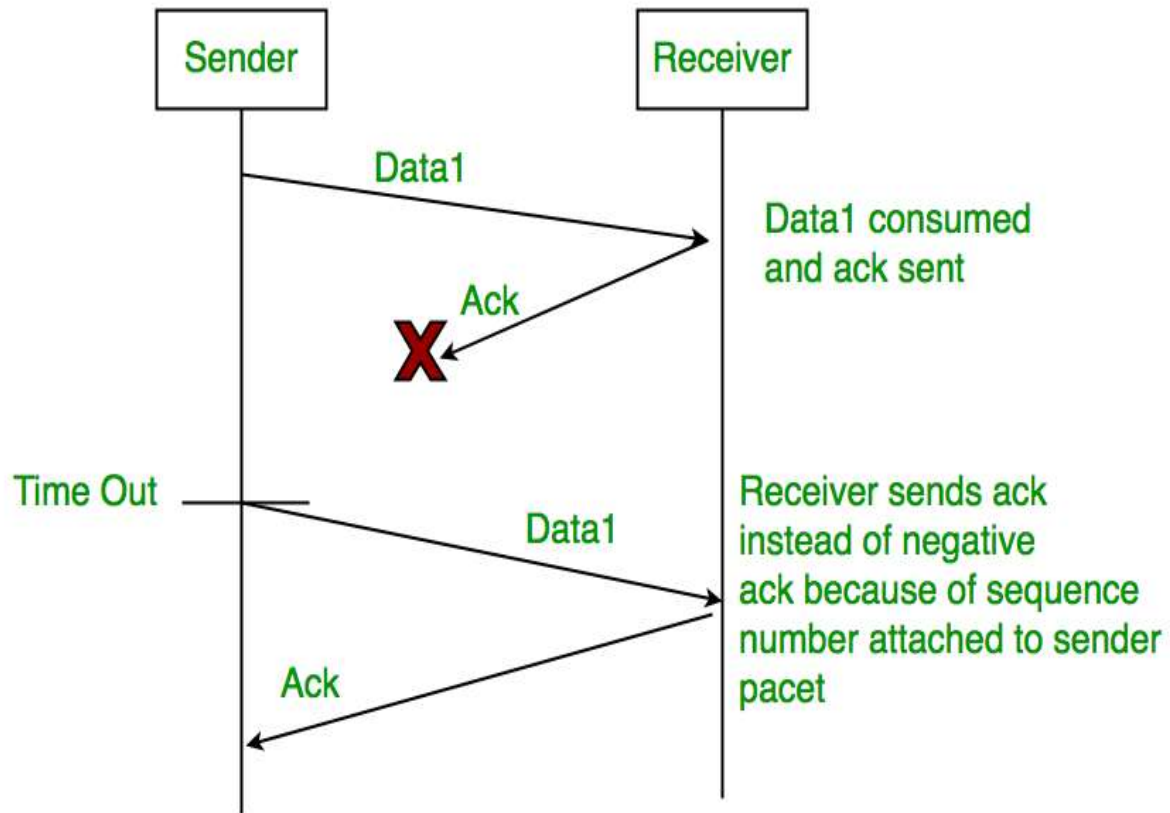
Time Out



Time Out

- Time out timer helps to solve the problem of **lost data packet**.
- After sending a data packet to the receiver, sender starts the time out timer.
- If the data packet gets acknowledged before the timer expires, sender stops the time out timer.
- If the timer **goes off** before receiving the acknowledgement, sender retransmits the **same data packet**.
- After retransmission, sender resets the timer.
- This prevents the occurrence of deadlock.

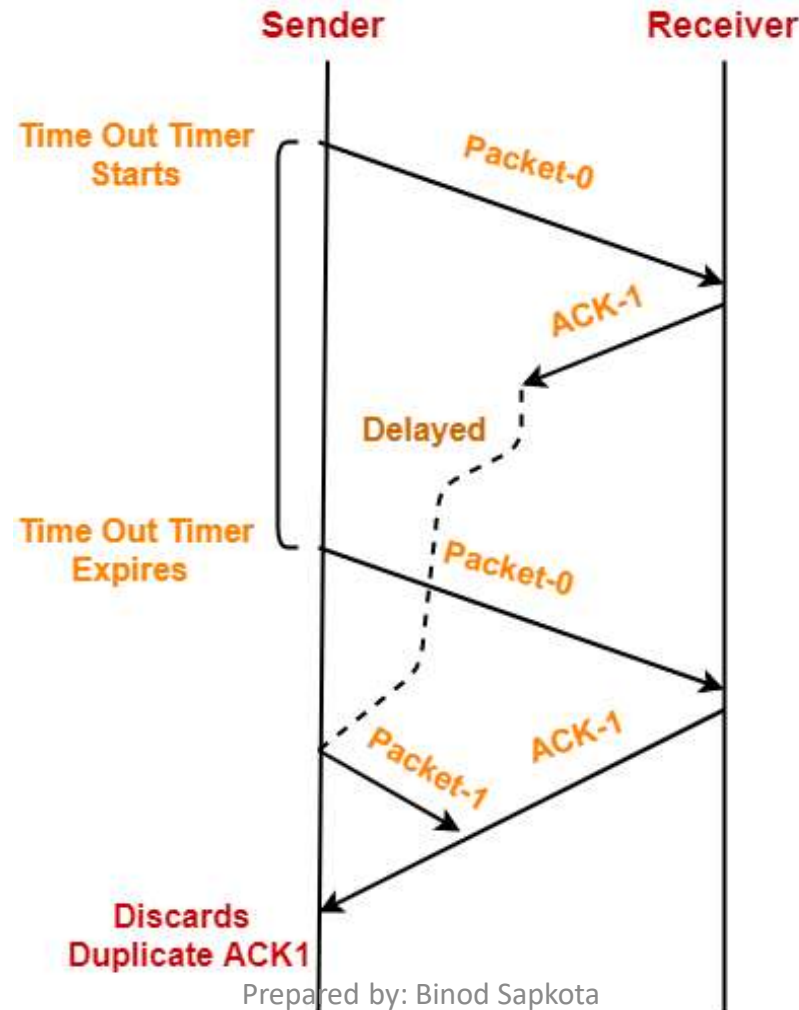
Sequence No.(on Data)



Sequence No.(on Data)

- Sequence number on data packets help to solve the problem of lost acknowledgement.
- Consider the acknowledgement sent by the receiver gets lost.
- Then, sender retransmits the same data packet after its timer goes off.
- This prevents the occurrence of deadlock.
- The sequence number on the data packet helps the receiver to identify the duplicate data packet.
- Receiver discards the duplicate packet and re-sends the same acknowledgement.

Sequence No.(on Acknowledgement)



Sequence No.(on Acknowledgement)

- Sequence number **on acknowledgements** help to solve the problem of **delayed acknowledgement**.
- Sender sends a data packet with sequence number-0 to the receiver.
- Receiver receives the data packet correctly.
- Receiver now expects data packet with sequence number-1.
- Receiver sends the acknowledgement ACK-1.
- Acknowledgement ACK-1 sent by the receiver gets delayed in reaching the sender.
- Sender receives no acknowledgement and time out occurs.

Sequence No.(on Acknowledgement)

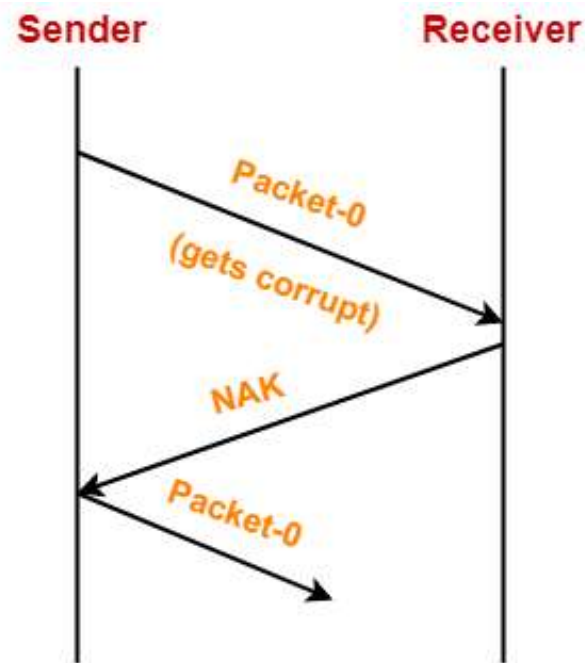
- Sender retransmits the same data packet with sequence number-0.
- This will be a duplicate packet for the receiver.
- Receiver receives the data packet and discovers it is the duplicate packet.
- It expects the data packet with sequence number-1 but receiving the data packet with sequence number-0.
- It discards the duplicate data packet and re-sends acknowledgement ACK-1.
- ACK-1 requests the sender to send a data packet with sequence number-1.

Sequence No.(on Acknowledgement)

- Two acknowledgements ACK1 reaches the sender.
- When first acknowledgement ACK1 reaches the sender, sender sends the next data packet with sequence number 1.
- When second acknowledgement ACK1 reaches the sender, sender **rejects the duplicate acknowledgement.**
- This is because it has already sent the data packet with sequence number-1 and now sender expects the acknowledgement with sequence number 0 from the receiver.

Problem of Damaged Packet

- If receiver receives a corrupted data packet from the sender, it sends a negative acknowledgement (NAK) to the sender.
- NAK requests the sender to send the data packet again.



Stop and Wait Protocol Vs Stop and Wait ARQ

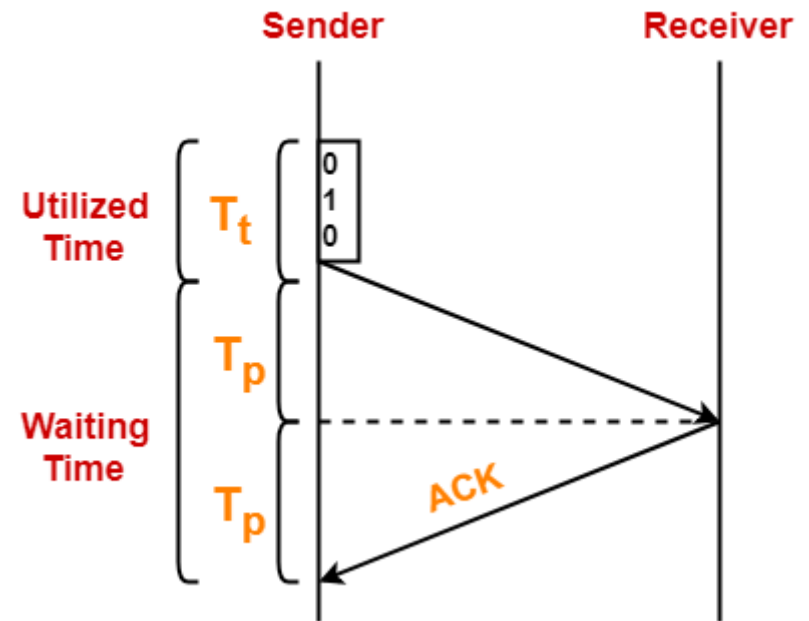
Stop and Wait Protocol	Stop and Wait ARQ
It assumes that the communication channel is perfect and noise free.	It assumes that the communication channel is imperfect and noisy.
Data packet sent by the sender can never get corrupt.	Data packet sent by the sender may get corrupt.
There is no concept of negative acknowledgements.	A negative acknowledgement is sent by the receiver if the data packet is found to be corrupt.
There is no concept of time out timer.	Sender starts the time out timer after sending the data packet.
There is no concept of sequence numbers.	Data packets and acknowledgements are numbered using sequence numbers.

Limitation of Stop and Wait ARQ

- The major limitation of Stop and Wait ARQ is its very less efficiency.
- In stop and wait ARQ,
- Sender window size is 1.
- This allows the sender to keep only one frame unacknowledged.
- So, sender sends one frame and then waits until the sent frame gets acknowledged.
- After receiving the acknowledgement from the receiver, sender sends the next frame.

Limitation of Stop and Wait ARQ

- Here,
- Sender uses T_t packet transmission time for transmitting the packet over the link.
- Then, sender waits for $2 \times T_p$ time.
- After $2 \times T_p$ time, sender receives the acknowledgement for the sent frame from the receiver.
- Then, sender sends the next frame.
- This $2 \times T_p$ waiting time is the actual cause of less efficiency.
- Efficiency(η)=useful time (T_t/T_{total})= $T_t/(T_t+2T_p)=1/(1+2a)$
- Where $a=T_p/T_t$
- $\eta \geq 0.5$



Useful Terms:

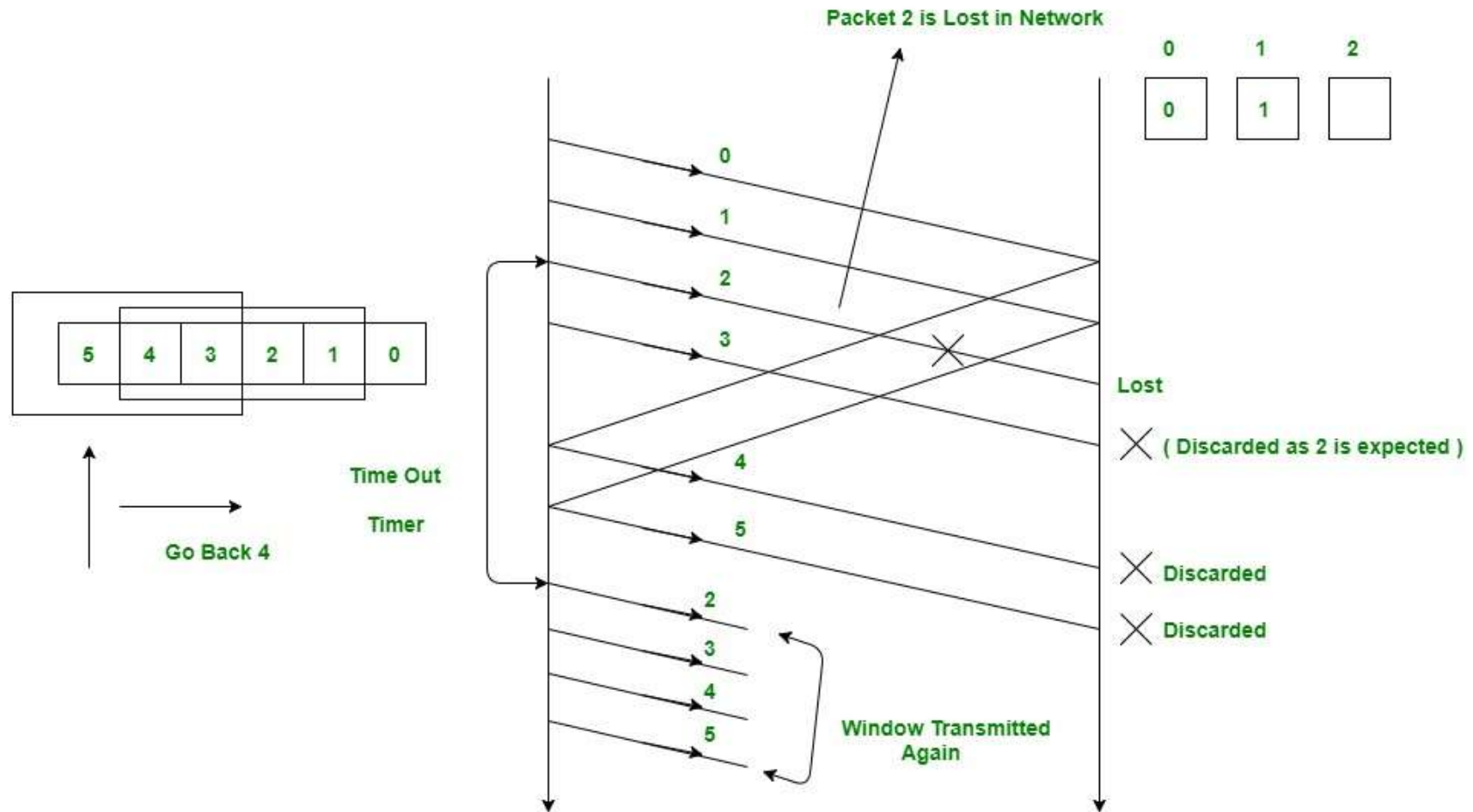
- **Propagation Delay:** Amount of time taken by a packet to make a physical journey from one router to another router or sender to receiver.
- Propagation Delay = (Distance between routers) / (Velocity of propagation) = d/v
- Round Trip Time (**RTT**) = $2 * \text{Propagation Delay}(T_p)$
- Time Out (**TO**) = $2 * \text{RTT}$
- Time To Live (**TTL**) = $2 * \text{Time Out}$. (Maximum TTL is 180 seconds)
- Effective Bandwidth or Throughput = $\eta * B$;
- $T_t = \text{length of packet} / \text{Bandwidth} = L/B$
- $\eta = 1/(1+2a) = 1/(1+2 * T_p/T_t) = 1/(1+2 * d/v * B/L)$; $w = \text{max of } (1+2a) \text{ packets}$
- If $d \uparrow$ - $\eta \downarrow$ so Stop and wait is for LAN no good for WAN
- If $L \uparrow$ - $\eta \uparrow$ so bigger packet is good

Efficiency Improvement

- The efficiency of stop and wait ARQ can be improved by increasing the **window size**.
- This allows the sender to keep more than one unacknowledged frame in its window.
- Thus, sender can send frames in the waiting time too.

Go Back N

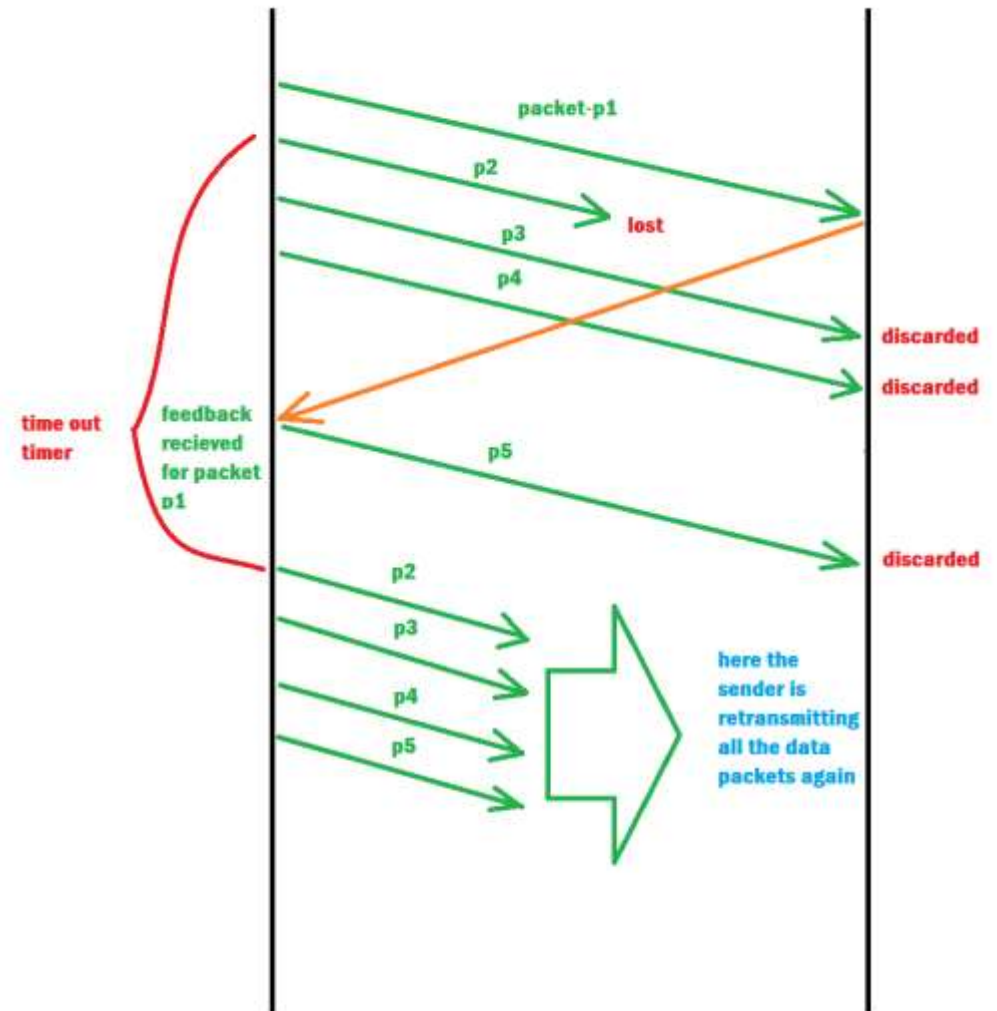
- The sender sends N packets which is equal to the window size. Once the entire window is sent, the sender then waits for a **cumulative ACK** to send more packets.
- On the receiver end, it receives only in-order packets and discards out-of-order packets. As in case of packet loss, the entire window would be re-transmitted.
- In Go back N, sender window size is **N(>1)** and receiver window size is always **1** for any value of **N**.
- If Sender window size = N. Example in Go back 10, sender window size will be 10.
- Go back N uses **cumulative acknowledgements**.
- One acknowledgement is used for many packets.
 - The main advantage is traffic is less.
 - A disadvantage is less reliability as if one ack is the loss that would mean that all the packets sent are lost.



GBN Example...

- Assume, We have sender window size of 4.
- Now the sender has sent the packets 0, 1, 2 and 3.
- After acknowledging the packets 0 and 1, receiver is now expecting packet 2 and sender window has also slid to further transmit the packets 4 and 5.
- Now suppose the packet 2 is lost in the network, Receiver will discard all the out of order packets (3,4,5) which sender has transmitted after packet 2 as it is expecting sequence number of 2.
- On the sender side for every packet send there is a time out timer which will expire for packet number 2.
- Now from the last transmitted packet 5 sender will go back to the packet number 2 in the current window and transmit all the packets till packet number 5 [2,3,4,5].
- That's why it is called Go Back N.
- Go back means sender has to go back N places from the last transmitted packet [5] in the unacknowledged window and but not from the point where the packet is lost.[2]

Consider a sender has to send data packets indexing from **p1 to p4**, it sends all the data packets in order (from p1 to p4), but the receiver has only received p1 and the data packet p2 is lost somewhere in the network, then the receiver declines all the data packets after p2 (i.e. p3, p4,p5) because the receiver is waiting for packet p2 and will not accept any other data packet than that. So, now as the time out time index of p2 expires, the sender goes back 3 packets and starts sending all the data packets from p2 to p5 again.



Go Back N

- Efficiency = N/W = Sender Window Size in Protocol / $(1 + 2a)$
- Efficiency of Go back N = $N / (1 + 2a)$ $Ws(=N)$: sender window size
- 1. If $T_t=1\text{msec}$, $T_p=49.5\text{msec}$ for GB10, $\eta=?$
 - $\eta=N/W=N / (1 + 2a)=10/100=10\%$;
 - W =maximum sender window capacity=max of $(1+2a)$ packets could send
- If $B=40\text{mbps}$, Throughput or Bandwidth utilization=?
 - $Th = \eta * B = 10/100 * 40 = 4\text{mbps}$ (effectively used)
- Exercise: In GB3, if every 5th packet is being transmitted is lost and if we have to send 10 packets, then how many transmissions are required?
- <https://www.youtube.com/watch?v=ZLtkhsgQp8U>

*Go Back N

- IN GB4 (entire window 4/3), if every 6/5th packet is being transmitted is lost and no ACK from the receiver is lost; if we have to send 10/9 packets, then how many transmissions are required?
 - 1 2 3 4 5 6 7 8 9 10
(6th) ↑
 - 1 2 3 4 5 6 7 8 9 6 7 8 9 10 8 9 10 (whether 7 8 9 is accepted or rejected, it is transmitted)
(6th) ↑
 - transmission time is now= 1 2 3 4 5 6 7 8 9 6 7 8 9 10 8 9 10 = 17 (10+7)
 - Actual transmission=10, extra transmission = 7
 - 16 transmission for GB3 extra 6

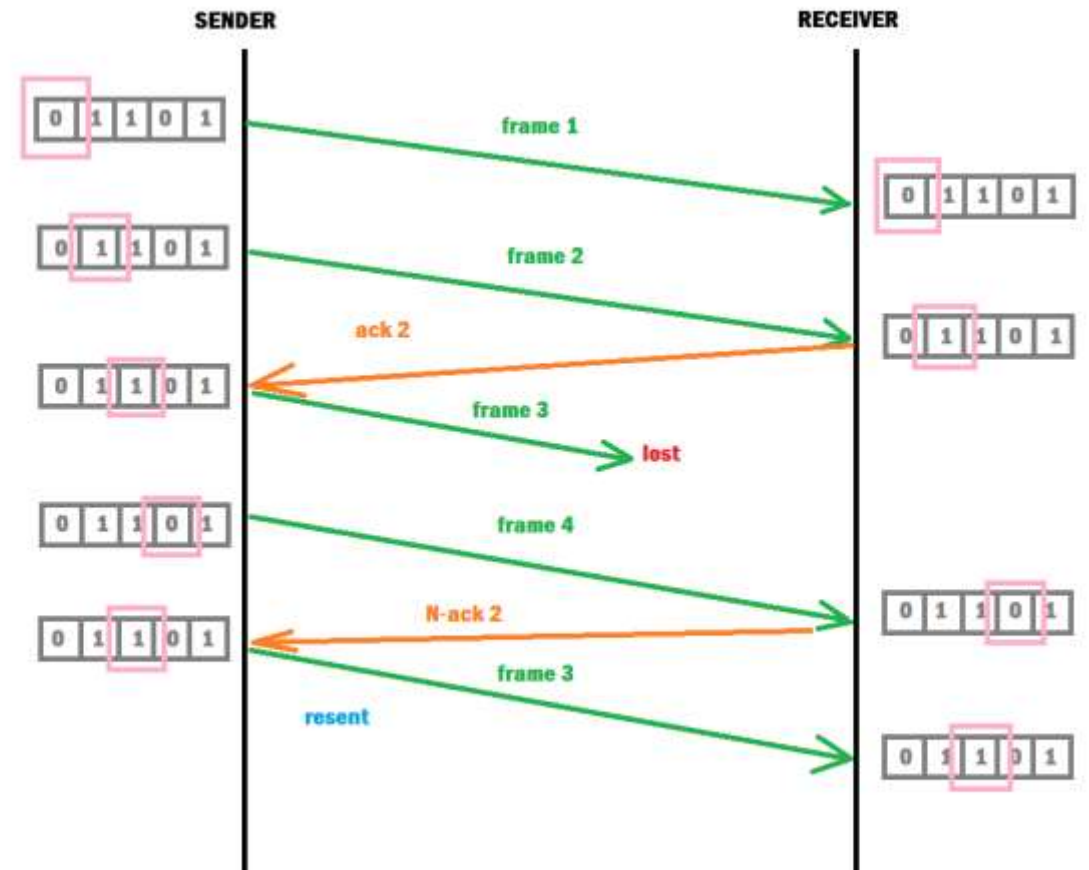
Selective Repeat

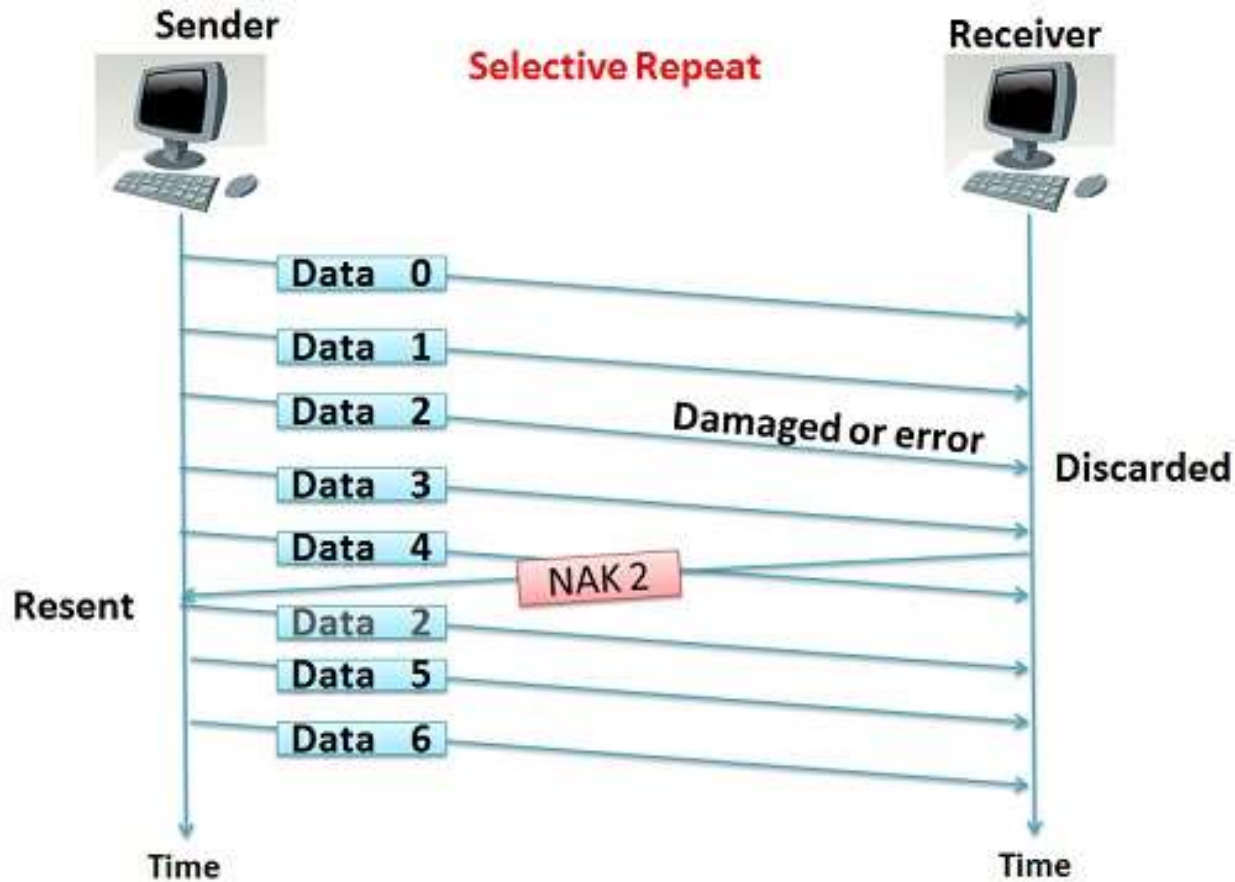
- The sender sends packet of window size N and the receiver acknowledges all packet whether they were received in order or not.
- In this case, the receiver maintains a buffer to contain **out-of-order** packets and sorts them.
- The sender selectively re-transmits the lost packet and moves the window forward.
- Sender's Windows (W_s) = Receiver's Windows (W_r)
- SRP also requires full duplex link.
- backward acknowledgements are also in progress.

Selective Repeat

- SR protocol requires sorting at the receiver's side and searching at the sender's side.
 - Receiver does not reject the out of order frames.
 - Receiver accepts the out of order frames and sort them later.
 - Thus, only the missing frame has to be sent by the sender.
 - For sending the missing frame, sender performs searching and finds the missing frame.
 - Then, sender selectively repeats that frame.
 - Thus, only the **selected frame is repeated and not the entire window**.
 - That is why, the protocol has been named as **"Selective Repeat Protocol"**.

the sender first sends **frame1** (i.e.0) and immediately the receiver acknowledges the frame and then, the **frame2** is transmitted which too got acknowledged. Now, as we can see **frame3** is lost during the transmission & was unable to reach the receiver, but the next frame, which is **frame4** is transmitted and also accepted by the receiver, who then sends a negative acknowledgment due to which the sender only transmits **frame3** again instead of retransmitting all the frames starting from frame3(which happens in Go-Back-N ARQ).





Efficiency = $N / (1 + 2a)$ Where a = Propagation delay / Transmission time(or delay)

Buffers = $N + N$

Sequence number = $N(\text{sender side}) + N(\text{Receiver Side})$

*Selective Repeat

- SR protocol leads to retransmission of lost frames after expiry of time out timer.
 - Consider a frame being sent to the receiver is lost on the way.
 - Then, it is retransmitted only after time out timer expires for that frame at sender's side.
- If we have $W_S=W_R=3$ (*irrelevant), 10 packets and 5th packet is lost then how many packets or total transmission we get?
- 1 2 3 4 5 6 7 8 9 10
- 1 2 3 4 5 5 6 7 8 9 9 10
- Only one packet is repeated not the entire window as in GBN.
- Total transmission=12 (actual = 10 and retrans =2)
- Solve: If 4th packet lost, total transmission??? (Ans: 13)
- But in GBN total transmission =18 (10+8) for this questions.
- <https://www.youtube.com/watch?v=Oipm5DdYYAs>

S.NO	GO-BACK-N PROTOCOL	SELECTIVE REPEAT PROTOCOL
1.	In Go-Back-N Protocol, if the sent frame are find suspected then all the frames are re-transmitted from the lost packet to the last packet transmitted.	In selective Repeat protocol, only those frames are re-transmitted which are found suspected.
2.	Sender window size of Go-Back-N Protocol is N.	Sender window size of selective Repeat protocol is also N.
3.	Receiver window size of Go-Back-N Protocol is 1.	Receiver window size of selective Repeat protocol is N.
4.	Go-Back-N Protocol is less complex.	Selective Repeat protocol is more complex.
5.	In Go-Back-N Protocol, neither sender nor at receiver need sorting.	In selective Repeat protocol, receiver side needs sorting to sort the frames.
6.	Efficiency of Go-Back-N Protocol is $N/(1+2*a)$	Efficiency of selective Repeat protocol is also $N/(1+2*a)$
7.	In Go-Back-N Protocol, type of Acknowledgement is cumulative.	In selective Repeat protocol, type of Acknowledgement is individual.

PROPERTIES	STOP AND WAIT	GO BACK N	SELECTIVE REPEAT
Sender window size	1	N	N
Receiver Window size	1	1	N
Minimum Sequence number	2	N+1	2N
Efficiency	$1/(1+2*a)$	$N/(1+2*a)$	$N/(1+2*a)$
Type of Acknowledgement	Individual	Cumulative	Individual
Supported order at Receiving end	—	In-order delivery only	Out-of-order delivery as well
Number of retransmissions in case of packet drop	1	N	1

Advantages depend on context either GBN or SR is used

Parameters	Stop and Wait	GBN	SR	*Remarks
Efficiency	$1/1+2a$	$N/1+2a$	$N/1+2a$	
Buffers	1+1	N+1	N+N	memory
sequence numbers	1=1	N+1	N+N	Minimum different
Retransmission (assume 1 pkt is lost)	1	N	1	
BW	Low	High	Moderate	
CPU required	low	moderate	high	Out of order packet Sorting at Rx and Searching at Tx
Implementation	low	moderate	high	

3.3 Examples of Data Link Protocol, HDLC, PPP

- **HDLC (High-level Data Link Control)** is a WAN protocol intended to perform the encapsulation of the data in the data link layer.
- The encapsulation of the data means to change the format of the data.
 - SDLC is the predecessor of the HDLC which stands for the **Synchronous Data Link Control protocol**.
 - Both SDLC and HDLC protocol are developed by IBM and submitted to the ANSI and ISO for the acceptance as the international standards.
- There are different networks exist in WAN. They use different encapsulation protocols. For example,
 - leased line (between two routers) and circuit switched network use **HDLC, PPP** and SLIP protocols.
 - Packet switched networks use **X.25, Frame relay** and **ATM** protocols.

HDLC (High-level Data Link Control)

- Frame format for the bit-oriented protocols



Frame format for HDLC protocol

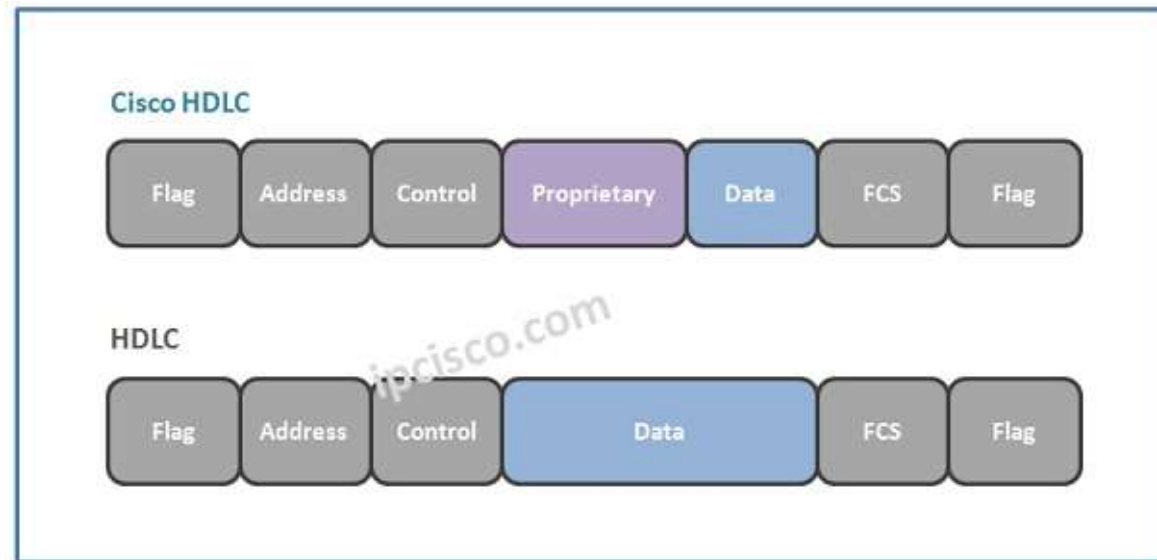
- **Address field** – It is used to describe the terminal.
- **Control field** – The bits in the control field is intended for the sequence number and acknowledgements. Control field indicates purpose or function of the HDLC Frame.
- **Data field** – This field is used to hold the information.
- **Checksum field** -In this field, the bits are reserved for the performing the cyclic redundancy code, used for error detection.
- HDLC frame consists of start and end flags with pattern 01111110.

HDLC...

- As HDLC uses standard pattern i.e. 01111110 to mark start and end of the frame, it may often confuse the receiver and hence to avoid this transparency problem bit stuffing is done.
 - In **bit stuffing**, 0 is added after every five 1s at the transmit side and 0 is deleted at the receive side.
- Broadly HDLC is a reliable protocol (uses **selective repeat** or **go-back-N**).
- It uses **full duplex** communication. It has **flow control mechanism** as it adjusts window size based on capability of the receiver.
- It uses physical layer clocking and synchronization to send and receive frames.

HDLC two Versions

- **HDLC** has two versions.
- One of them is the **standard one** and the other is the **Cisco proprietary version**.
 - The frame of standard version and Cisco proprietary version is similar.
 - Only in Cisco proprietary HDLC, there is one additional proprietary field.



RFC 1661-PPP (Point-to-Point Protocol)

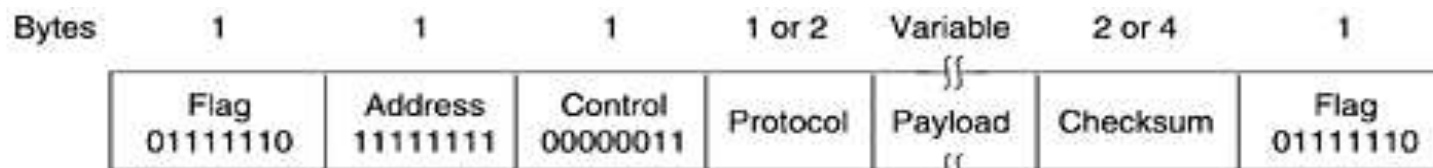
- **PPP** is also a WAN protocol, but there are several enhancements made in the PPP protocol after HDLC.
- Priorly, the PPP protocol is **not proprietary**, which means that it can be used with two different type of devices without committing changes over the format of the data.
- The main elements of PPP are – a mechanism for encapsulating multi-protocol datagrams, **LCP (Link Control Protocol)** and a group of **NCP (Network Control Protocols)**.
- LCP mainly **sets up, configure and test** the connections while NCP is responsible for **establishing and configuring the distinct network layer protocols**.

Frame format for the PPP

- Important fields in the PPP header and trailer include
 - **Flag**, 1 byte delimiter field
 - Address, 1 byte
 - Control, 1 byte
 - **Protocol identifier**, 2 byte field
 - Data payload, variable
 - **Frame Check Sequence (FCS)**, 2 byte field



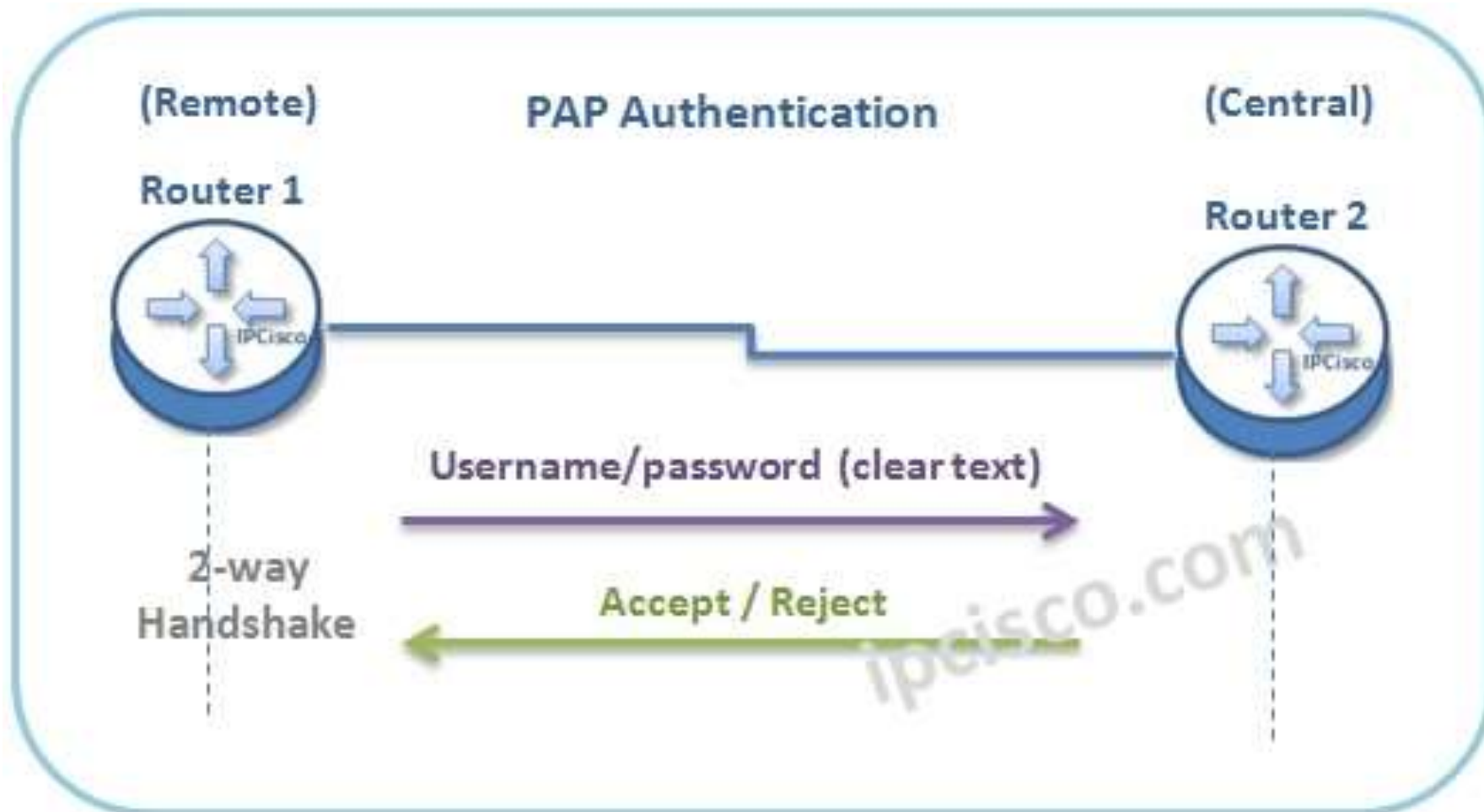
Frame format for PPP protocol



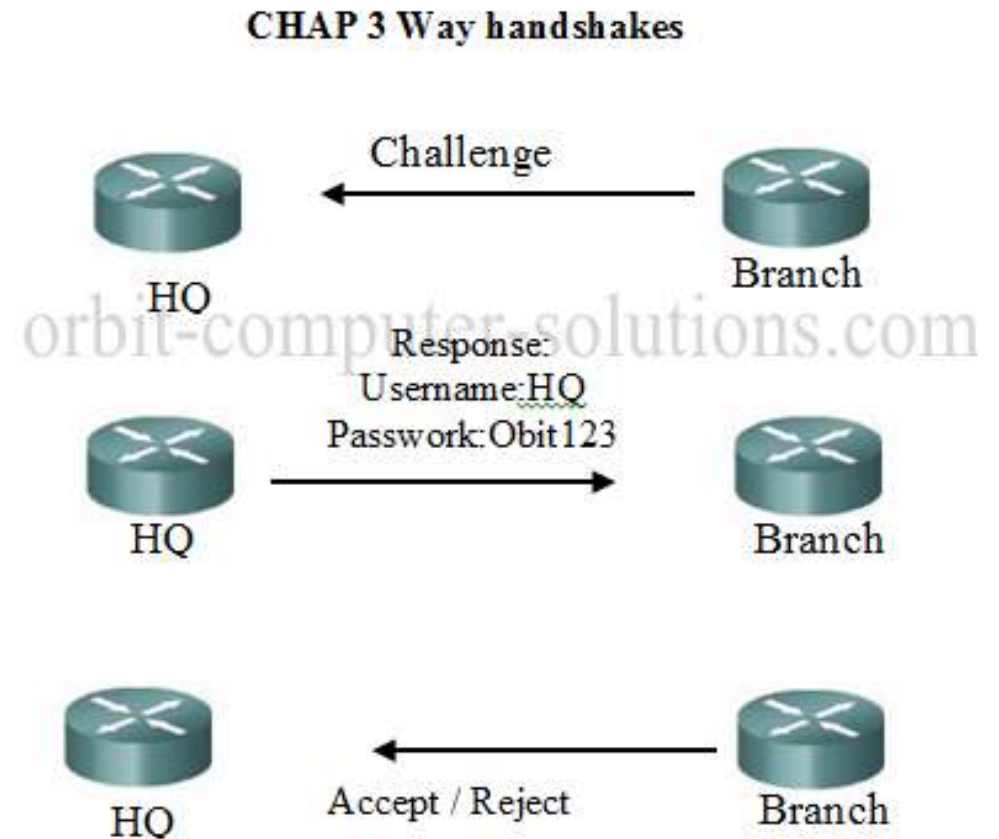
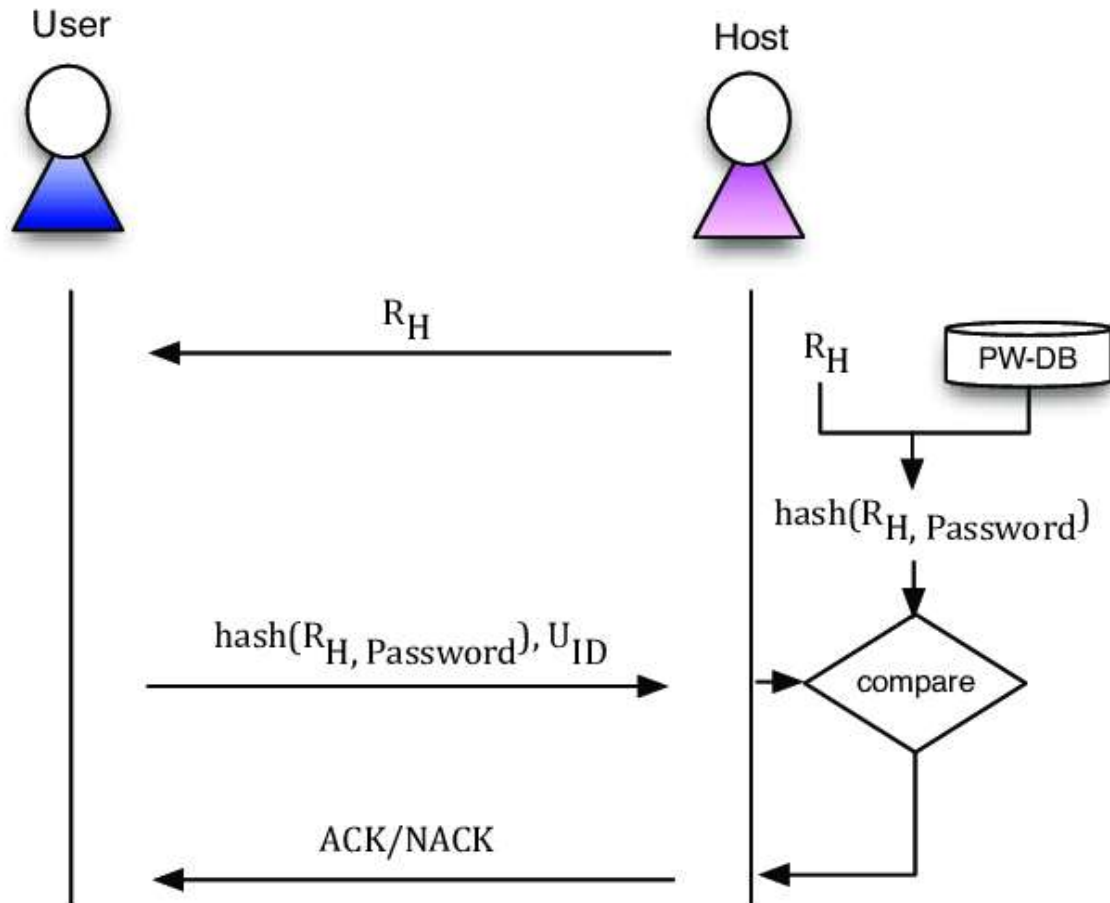
PPP Authentication

- **PPP** supports two Authentication Protocols.
 - **PAP (Password Authentication Protocol)**
 - **CHAP (Challenge Handshake Authentication Protocol)**
- **PAP** is the simplest Authentication method.
 - It uses 2-way handshake.
 - Both end send the passwords in “**clear text**” in this method.
 - And passwords are exchanged only at the beginning.
- **CHAP** is the more complex Authentication method.
 - CHAP uses 3-way handshake and with this mechanism it checks the remote node periodically.
 - CHAP uses **MD5** hash.
 - One end sends “**Hash**” to other node and the other node also sends a **hash**.
 - If the hashes are same, then the communication starts.

PAP (Password Authentication Protocol)



CHAP (Challenge Handshake Authentication Protocol)



How CHAP Works

- After the **PPP link encapsulation** phase is complete, the local router sends a **challenge message** to the remote host.
- The remote host sends a response with a value calculated using a one-way hash function, which is normally **Message Digest 5 (MD5)** based on the **password and challenge message**.
- The local router checks the **response from the remote host** against **its own** calculation of the expected hash value.
 - If there is a match, the initiating host **acknowledges** the authentication.
 - If the values don't match, it immediately terminates the connection.

CHAP...

- The Branch Router initiates the 3-way handshake and sends a **challenge message** to router HQ.
- HQ response to Branch's CHAP challenge by sending its **username and password**.
- Branch checks HQ's username and password in its local database for a possible match,
 - if there is a match, it accepts the connection. If not, it rejects.
- **Advantage:** CHAP provides protection against playback attack by using different challenge value that is unique and comes in random.
 - Because the challenge is unique and unpredictable, the resulting hash value is also unique and random. Which makes it difficult for 'guessing'.
- CHAP is more secure than PAP.
 - During link establishment, CHAP conducts periodic challenges to make sure that the remote host still has a valid password value.
 - While PAP basically stops working once authentication is established, this leaves the network vulnerable to attack.

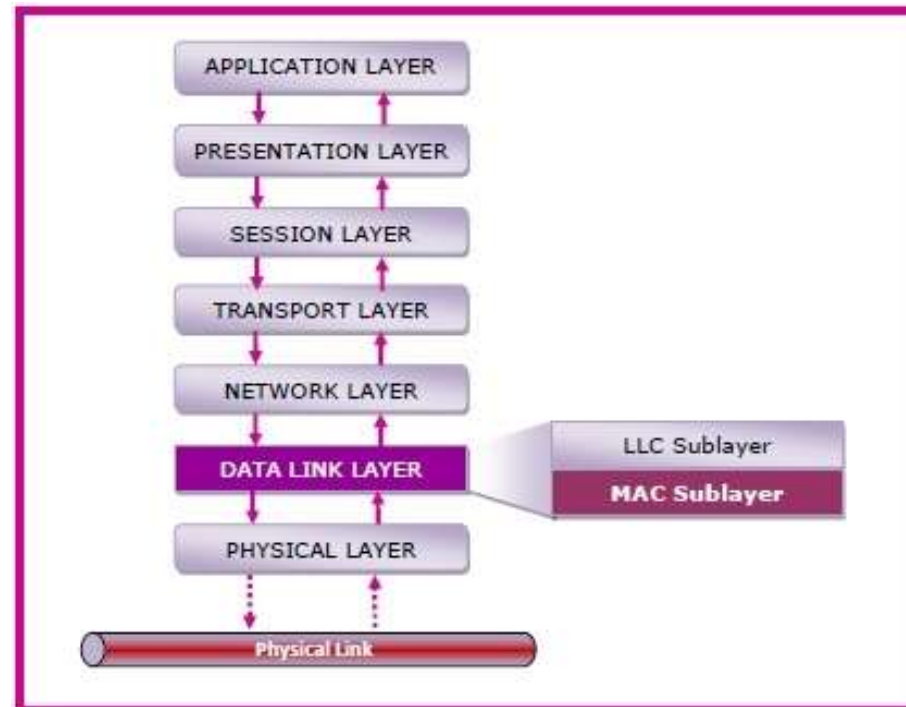
Comparison Chart

BASIS FOR COMPARISON	HDLC	PPP
Expands to	High-level Data Link Layer Protocol	Point-to-Point Protocol
Type of protocols	Bit-oriented protocol	Byte oriented protocol
Used in	Only synchronous media	Synchronous as well as asynchronous media
Authentication	No provision of authentication	Provides authentication
Dynamic addressing	Does not offer dynamic addressing.	Dynamic addressing is used.
Implemented in	Point-to-point and multipoint configurations.	Only point-to-point configurations.
Compatibility with other protocols	Can not be operated with non-Cisco devices.	Interoperable with non-Cisco devices also.

Prepared by: Binod Sapkota

3.4 The Medium Access Sub-layer

- The data Link layer is divided into two sublayers –
 - The logical link control (LLC) sublayer
 - The medium access control (MAC) sublayer



MAC

- It is responsible for flow control and multiplexing for transmission medium.
- It controls the transmission of data packets via remotely shared channels, provides addressing and channel access control mechanisms that enable several terminals or network nodes to communicate in a network.
- The MAC sublayer acts as an interface between the logical link control (LLC) Ethernet sublayer and Layer 1 (the physical layer).
- The MAC sublayer emulates a full-duplex logical communication channel in a multipoint network.
- The MAC sublayer uses **MAC protocols** to prevent collisions.

What MAC layer does?

- (ON SENDERS END) It receives packet from network layer and attach header to it, header includes **48-bit MAC address** (*which is unique to each machine provided by manufacturer unlike IP address which is provided by network provider*), also includes **Error correction bits**
- (ON RECIVERS END) It receives as a frame from physical layer, verify if frame has destination MAC address of this computer only if yes accept that frame, removes frame header and forward remaining payload to Network layer
- It also ensures a collision free transmission of frames in some of the protocol like 802.3(Ethernet LAN) or 802.11(Wireless LAN)

MAC Addresses

- MAC address is a unique identifier allotted to a network interface controller (NIC) of a device.
- It is used as a network address for data transmission within a network segment like Ethernet, Wi-Fi, and Bluetooth.
- MAC address is assigned to a network adapter at the time of manufacturing.
 - It is hardwired or hard-coded in the network interface card (NIC).
- A MAC address comprises of six groups of two hexadecimal digits, separated by hyphens, colons, or no separators.
- An example of a MAC address is 00:0A:89:5B:F0:11.
 - Upper 3 bytes represents the OUI (Organization Unique Identifier) also called Manufacturer ID
 - Lower 3 bytes represent the Device ID
- Each computer has its own unique MAC address.

3.7 The channel allocation problem

- **Basic problem:** How to allocate a multiaccess channel among competing users.
- We need a set of rules (i.e. a protocol) to allow each user to communicate and avoid interference.
- The solutions to this problem that are used in practice are
 - **static**
 - With a static approach, the channel's capacity is essentially divided into fixed portions; each user is then allocated a portion for all time.
 - If the user has no traffic to use in its portion, then it goes unused.
 - **Dynamic**
 - With a dynamic approach the allocation of the channel changes based on the traffic generated by the users.
- Generally, a static allocation performs better when the traffic is predictable. A dynamic channel allocation tries to get better utilization and lower delay on a channel when the traffic is unpredictable.

Static Channel Allocation Techniques

- Two common static channel allocation techniques are **TDMA** and **FDMA**.
- Time Division Multiple Access (TDMA) –
 - With TDMA the time axis is divided into time slots of a fixed length.
 - Each user is allocated a fixed set of time slots at which it can transmit.
 - TDMA requires that users be synchronized to a common clock.
 - Typically extra overhead bits are required for synchronization.
- Frequency Division Multiple Access (FDMA) –
 - With FDMA the available frequency bandwidth is divided into disjoint frequency bands.
 - A fixed band is allocated to each user.
 - FDMA requires a guard band between user frequency bands to avoid cross-talk.

SCA...

- Another static allocation technique is Code Division Multiple Access (**CDMA**), this technique is used in many wireless networks
- The performance of static channel allocation depends on:
 - The variation in the number of users over time
 - The nature of the traffic sent by the user
- If the traffic on a shared medium is from a fixed number of sources each transmitting at a fixed rate, **static channel allocation** can be very efficient.
- When the traffic sent by a user is bursty, then, under a static allocation, a user's portion of the channel may be empty when another user could use it. This leads one to think that a **dynamic allocation** will perform better in such cases.

Dynamic Channel Allocation Techniques

- In dynamic channel allocation scheme, no user is assigned fixed frequency or fixed time slot.
- All users are dynamically assigned frequency or time slot, depending upon the requirements of the user.
- The allocation is done considering a number of parameters so that **transmission interference is minimized**.
- This allocation scheme **optimizes bandwidth usage** and **results is faster transmissions**.
- Dynamic channel allocation is further divided into **centralized** and **distributed** allocation.

DCA

- The dynamic channel allocation schemes can be divided into three categories –
- Interference Adaptive Dynamic Channel Allocation (IA-DCA)
- Location Adaptive Dynamic Channel Allocation (LA-DCA)
- Traffic Adaptive Dynamic Channel Allocation (TA-DCA)
- Advantages:
- Dynamic channel allocation schemes allots channels as needed. This results in optimum utilization of network resources. There are less chances of **denial of services** and **call blocking** in case of voice transmission. These schemes adjust bandwidth allotment according to traffic volume, and so are particularly suitable for bursty traffic.
- Disadvantages:
- Dynamic channel allocation schemes increases the computational as well as storage load on the system.

Hybrid Channel Allocation (HCA)

- Hybrid Channel Allocation is a combination of both Fixed Channel Allocation (FCA) and Dynamic Channel Allocation (DCA).
- The total number of channels or voice channels are divided into fixed and dynamic set.
- When a user make a call then first fixed set of channels are utilized but if all the fixed sets are busy then dynamic sets are used.
- The main purpose of HCA is to work efficiently under heavy traffic and to maintain a minimum S/I.

FIXED CHANNEL ALLOCATION(FCA)	DYNAMIC CHANNEL ALLOCATION(DCA)
Fixed number of channels or voice channels are allocated to cells.	Fixed number of channels are not allocated to cells.
If all the channels are occupied and user make a call then the call will be blocked in FCA.	If all the channels are occupied and user make a call then Base Station(BS) request more channel to the Mobile Station Center(MSC).
Frequency reuse is maximum because cells channels are separated by minimum reuse distance.	Frequency reuse is not that maximum in DCA because of channel randomness allocation.
In FCA no such complex algorithms are used.	In DCA complex algorithms are used to decide which available channel is most efficient.
Fixed Channel Allocation Strategy is less costly than the DCA	Dynamic Channel Allocation Strategy is costly because lot of computation is required in real-time.
In FCA allocated channels remains to the cell, once the call is completed	In DCA once the call is completed then the channel or the voice channel return to the MSC.
Mobile Station Center(MSC) has less responsibilities.	Mobile Station Center(MSC) has more signaling load and responsibilities.

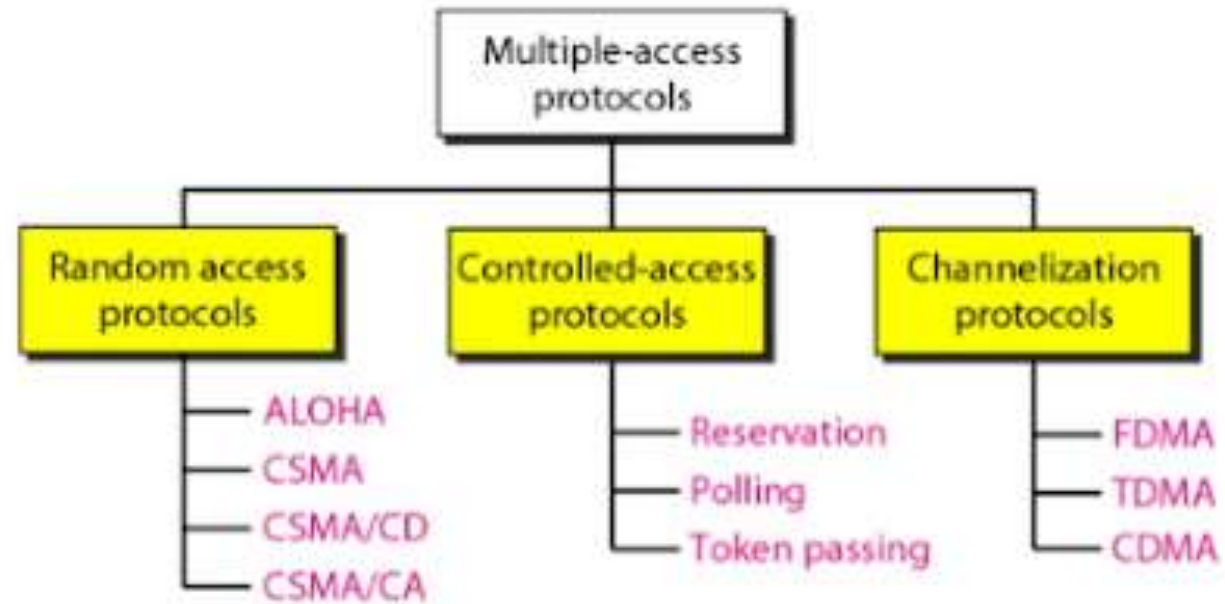
3.4 Multiple Access Protocols

- The Data Link Layer is responsible for transmission of data between two nodes. Its main functions are-
 - Data Link Control
 - Multiple Access Control
- **Data Link control –**
 - The data link control is responsible for reliable transmission of message over transmission channel by using techniques like framing, error control and flow control
- **Multiple Access Control –**
 - If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously.
 - Hence multiple access protocols are required to decrease collision and avoid crosstalk.

Multiple Access Links and Protocols

- Three types of links:
 - (a) Point-to-point (single wire) e.g. HDLC ,PPP
 - (b) Broadcast (shared wire or medium; eg, E-net, wireless, etc.)
 - Broadcast channels are often used in Local Area Networks (LANs)
 - (c) Switched (eg, switched E-net, ATM etc)
- MA (Multiple Access) indicates that many devices can connect to and share the same network.
 - All devices have equal access to use the network when it is clear.

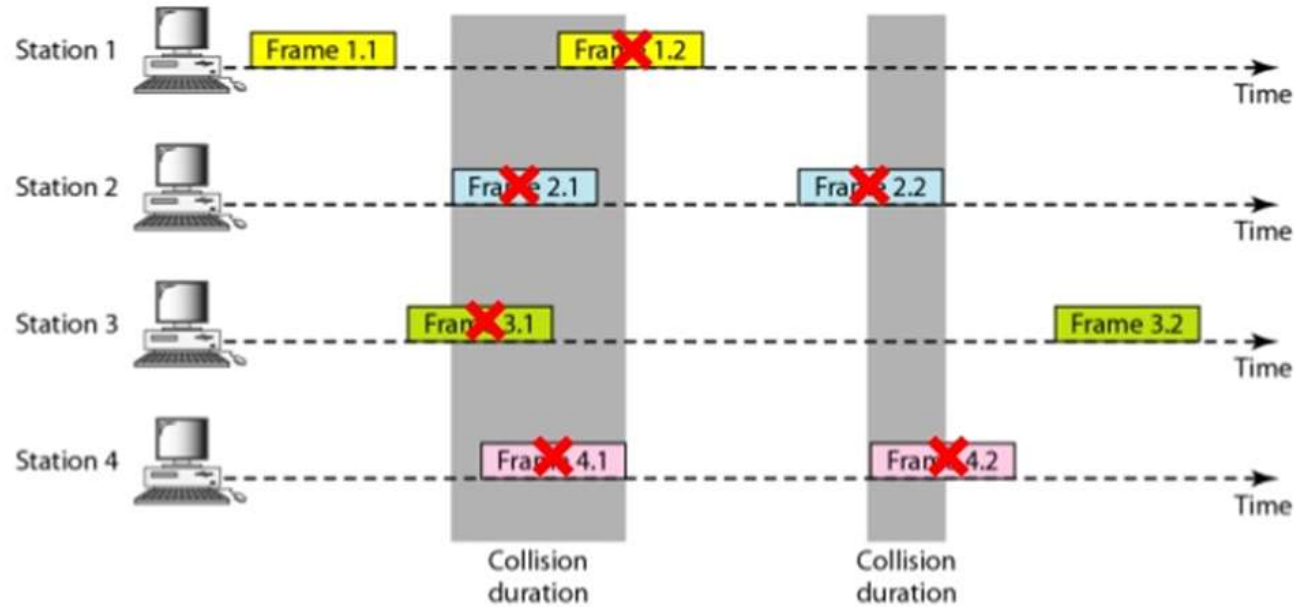
Multiple Access Protocols



3.4 Random Access Protocol

- In this, all stations have same superiority that is no station has more priority than another station.
- Any station can send data depending on medium's state(idle or busy).
- It has two features:
 - There is no fixed time for sending data
 - There is no fixed sequence of stations sending data

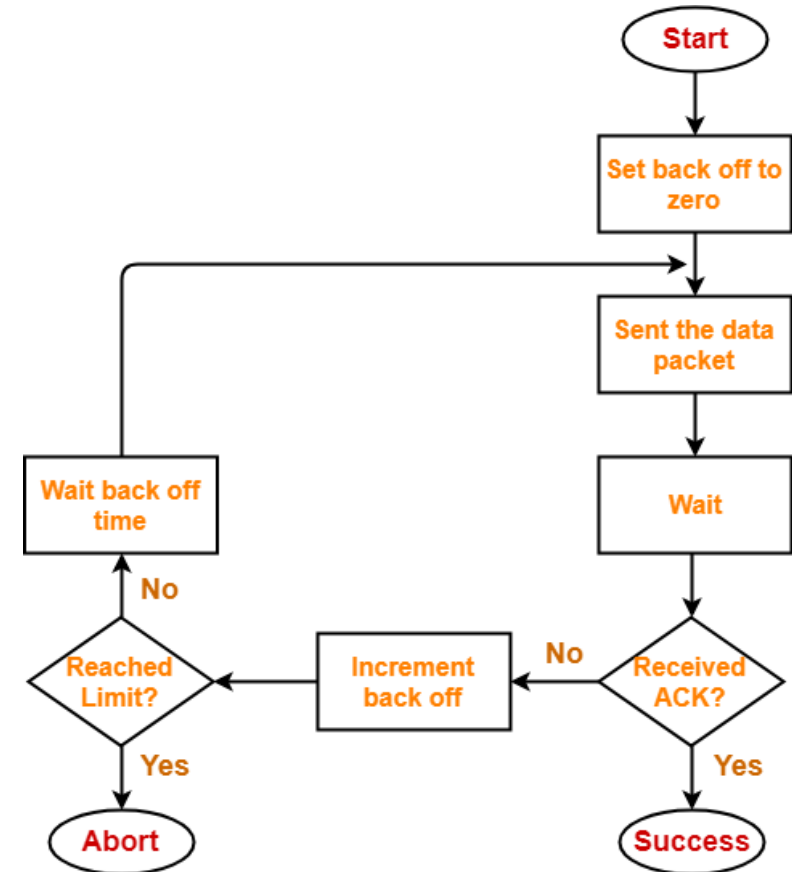
Frames in Pure ALOHA



15

Time and medium are common for all the stations

<https://www.youtube.com/watch?v=j4-r0e7DjqY>



Flowchart for Pure Aloha

Pure ALOHA

- It allows the stations to transmit data **at any time** whenever they want.
- After transmitting the data packet, station **waits for some time**.
- Transmitting station **receives an acknowledgement** from the receiving station. In this case, transmitting station assumes that the transmission is **successful**.
- Transmitting station **does not receive** any acknowledgement within specified time from the receiving station. In this case, transmitting station assumes that the transmission is **unsuccessful**.

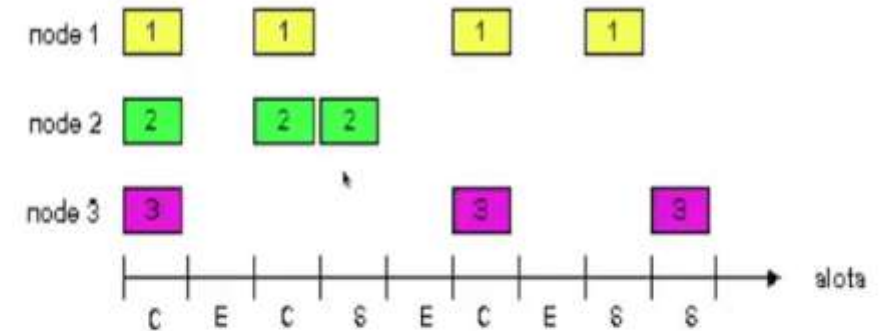
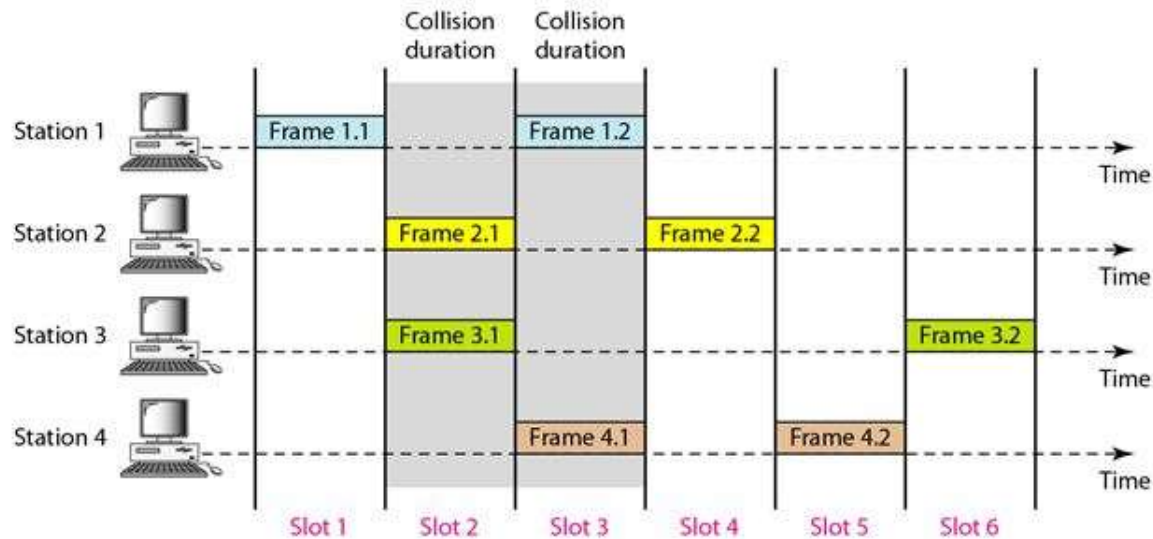
Pure ALOHA...

- Then, Transmitting station uses a (exponential) **Back Off Strategy** and waits for some **random amount of time/back off time**.
- After back-off time, it transmits the data packet **again**.
- It keeps trying until the **back off limit** is reached after which it **aborts** the transmission.
- Since the different stations wait for different amount of time, the probability of further collision decreases.
- The throughput of pure aloha is maximized when frames are of **uniform** length.
- If the first bit of the new frame overlaps with just the last bit of frame just finished ,both frame will be totally destroyed and both will have to be transmitted later.
- LAN based protocol (almost **obsolete**).

Pure ALOHA...

- Random Access Protocol
- Any time transmission
- Acknowledgement is there
- Only transmission time no propagation time
- Vulnerable time (V_T) = $2 * T_{fr}$ (frame transmission time) **condition for no collision**
- T_{fr} = length of message (L) / BW = 1000bits / 100kbps = 10ms (assume same)
- Efficiency, $\eta = G * e^{-2G}$ where G is the number of stations wish to transmit data ; **2** due to V_T
- Maximum efficiency, $d\eta/dG = 0$; $G = 1/2$ means half of the station (or 1 station from 2 station) transmission, efficiency will be maximum
- Putting $G = 1/2$; $\eta = 18.4\%$ means 18.4% of bandwidth is utilized rest of all collided.

Slotted Aloha...



Pros

- ▶ Single active node can continuously transmit at full rate of channel.
- ▶ Highly Decentralized
- ▶ Simple

Cons

- ▶ Collisions, Wasting Slots
- ▶ Idle Slots
- ▶ Nodes may be able to detect collisions in less than time to transmit packet
- ▶ Clock Synchronization

<https://www.youtube.com/watch?v=aqWTNk90zRA>

Slotted Aloha

- Slotted Aloha **divides the time of shared channel** into discrete intervals called as **time slots** (slot duration= T_t).
- Any station can transmit its data in any time slot.
- The only condition is that station must start its transmission from the **beginning of the time slot**.
- If the beginning of the slot is missed, then station has to wait until the beginning of the next time slot.
- A collision may occur if two or more stations try to transmit data at the beginning of the same time slot.
- Slotted ALOHA does not allow to transmit the data whenever the station has the data to be send.

Slotted Aloha

- Vulnerable time (V_T) = T_{fr} (frame transmission time) **condition for no collision**
- T_{fr} = length of message (L) / BW = 1000bits / 100kbps = 10ms (assume same)
- Efficiency, $\eta = G * e^{-G}$ where G is the number of stations wish to transmit data
- Maximum efficiency, $d\eta/dG = 0$; $G = 1$
- Putting $G = 1$; $\eta = 36.8\%$ (double than Pure Aloha), means One station gives maximum efficiency
- It means 36.8% of bandwidth is utilized rest of all bits are wasted due to collision. If BW = 100kbps then 36.8 kbps are transmitted.

Pure Aloha	Slotted Aloha
Any station can transmit the data at any time.	Any station can transmit the data at the beginning of any time slot.
The time is continuous and not globally synchronized.	The time is discrete and globally synchronized.
Vulnerable time in which collision may occur = $2 \times T_t$	Vulnerable time in which collision may occur = T_t
Probability of successful transmission of data packet = $G \times e^{-2G}$	Probability of successful transmission of data packet = $G \times e^{-G}$
Maximum efficiency = 18.4% (Occurs at $G = 1/2$)	Maximum efficiency = 36.8% (Occurs at $G = 1$)
The main advantage of pure aloha is its simplicity in implementation.	The main advantage of slotted aloha is that it reduces the number of collisions to half and doubles the efficiency of pure aloha.

Carrier Sense Multiple Access (CSMA)

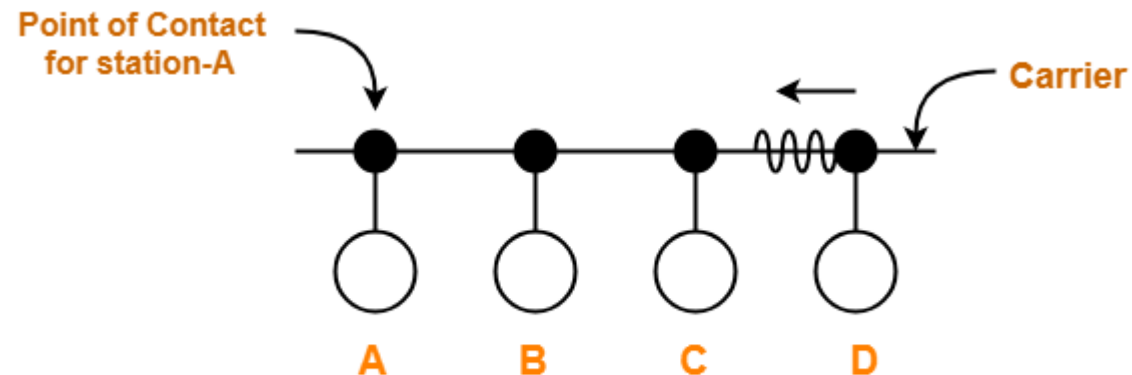
- CSMA is a network access method used on shared network topologies such as Ethernet **to control access to the network**.
- **Devices attached** to the network cable **listen** (carrier sense) before transmitting.
 - If the channel is in use, devices wait before transmitting.
 - (doesn't sense the entire /complete channel, but only **node's point of contact** .
- Even though devices attempt to sense whether the network is in use, there is a good chance that **two stations** will attempt to access it **at the same time**.
- **On large networks, the transmission time between one end of the cable and another is enough that one station may access the cable even though another has already just accessed it.**
- There are two methods for avoiding these so-called collisions,
 - CSMA/**CD** (collision detection)
 - CSMA/**CA** (Collision Avoidance)

CSMA/CD

- Any station, at any time willing to transmit the data senses the carrier
- If it finds the carrier free, it starts transmitting its data packet otherwise not.
- **How?**
- Each station can sense the carrier only at its point of contact with the carrier.
- It is not possible for any station to sense the entire carrier.
- Thus, there is a huge possibility that a station might sense the carrier/channel/medium free even when it is actually not.
- **No Ack system** .If so what method is used?

CSMA/CD...

- At the current instance,
- If station A senses the carrier at its point of contact, then it will find the carrier free.
- But the carrier is actually not free because station D is already transmitting its data.
- If station **A** starts transmitting its data now, then it might lead to a collision with the data transmitted by station **D**.



https://www.youtube.com/watch?v=v_z888gQWq0

(Bus Topology)

Detecting the Collision

- In CSMA / CD,
- It is the responsibility of the transmitting station to detect the collision.
- For **detecting the collision**, CSMA / CD implements the following condition.
 - This condition is followed by each station-
 - **Transmission delay $\geq 2 \times$ Propagation delay**
 - **Note: The node A must transmit continuously until the collision back or seen by A.**
- According to this condition,
- Each station must transmit the data packet of size whose transmission delay is at least twice its propagation delay.
- If the size of data packet (in bits) is smaller, then collision detection would **not** be possible.

CSMA / CD...

- Transmission time or delay = Length of data packet (L) / Bandwidth (B) $\geq 2 \times T_p$
- Propagation delay = Distance between the two stations (D) / Propagation speed (V)
- Thus, $L \geq 2 \times B \times T_p \geq 2 \times B \times D / V$

$$\text{Efficiency } (\eta) = \frac{T_t}{T_t + 6.44 \times T_p}$$

OR

$$\text{Efficiency } (\eta) = \frac{1}{1 + 6.44 \times a}, \text{ where } a = T_p / T_t$$

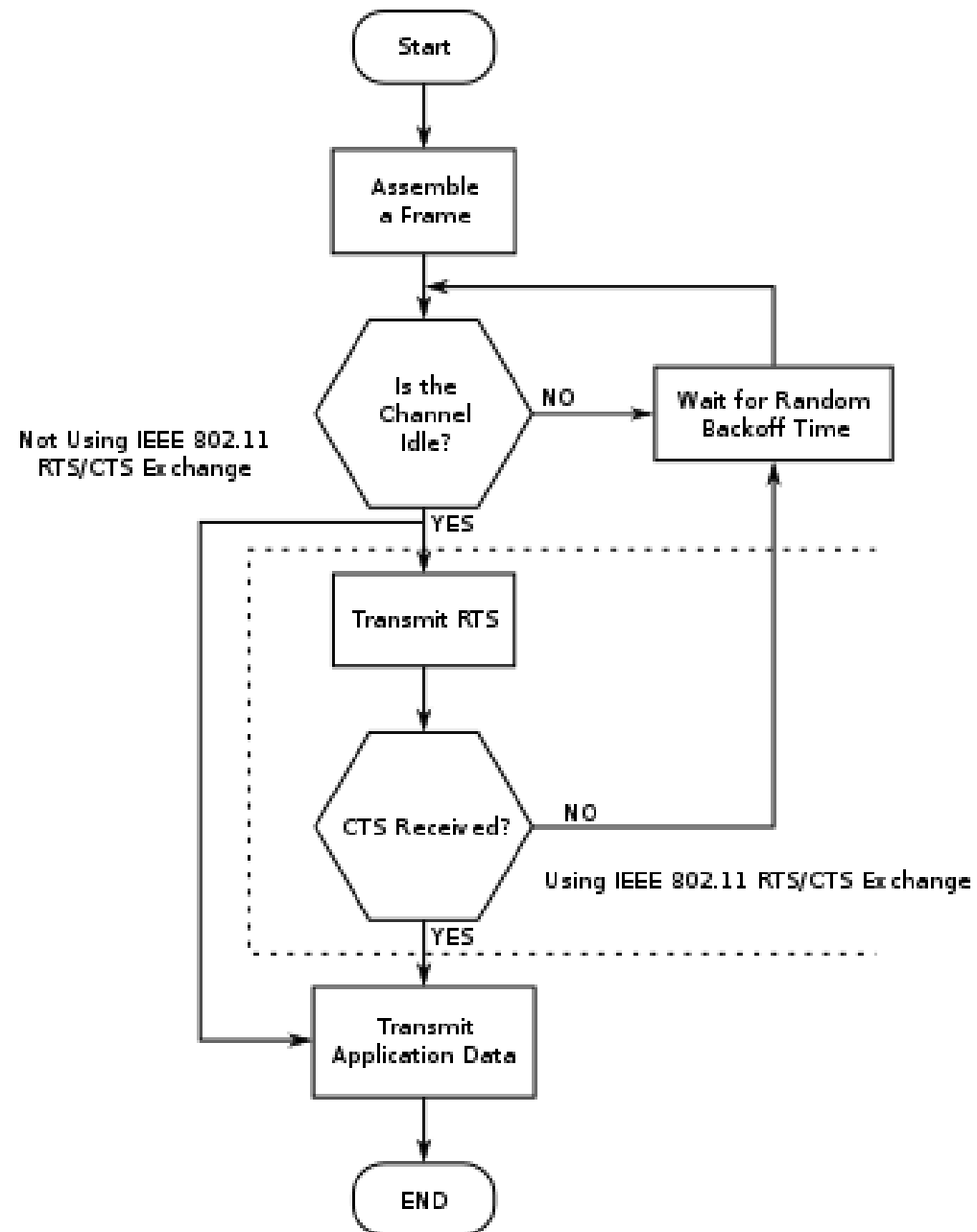
<https://www.youtube.com/watch?v=74zIRH-bj2c>

CSMA/CA (WLAN)

- Unlike CSMA/CD which deals with transmissions after collision has occurred, the nature of wireless medium makes it harder for a node to detect collisions. Therefore, CSMA/CA tries to prevent collisions before they happen.
- CA: The collisions are mostly avoided by waiting for a random amount of time where the channel is busy
- key idea:
- Sender and receiver exchange control frame (RTS/CTS) with each other before the sender actually transmits any data
- This exchange informs all nearby nodes that a transmission is about to begin

CSMA/CA

- Avoiding collision rather than allowing it to occur, is done using RTS/CTS signaling mechanism and NAV
- Sender node transmits short RTS(Request to Send) frame/packet to the receiver. It indicates duration of transmission in RTS signal (or how long the sender wants to hold the medium. Length of the data frame to be transmitted) using NAV (Network Allocation Vector). NAV is broadcasted to all the stations.
- Receiver node replies with short CTS (Clear to Send) packet. **It also indicates possible hidden nodes to the sender station/node.**
- Hidden nodes will not transmit for specified duration (i.e. NAV). In this way collision is avoided. Moreover as RTS and CTS is of short durations, collisions are less likely.
- NAV is used at MAC layer and widely used in IEEE 802.11 (WLAN) and 802.16 (WiMAX) standards



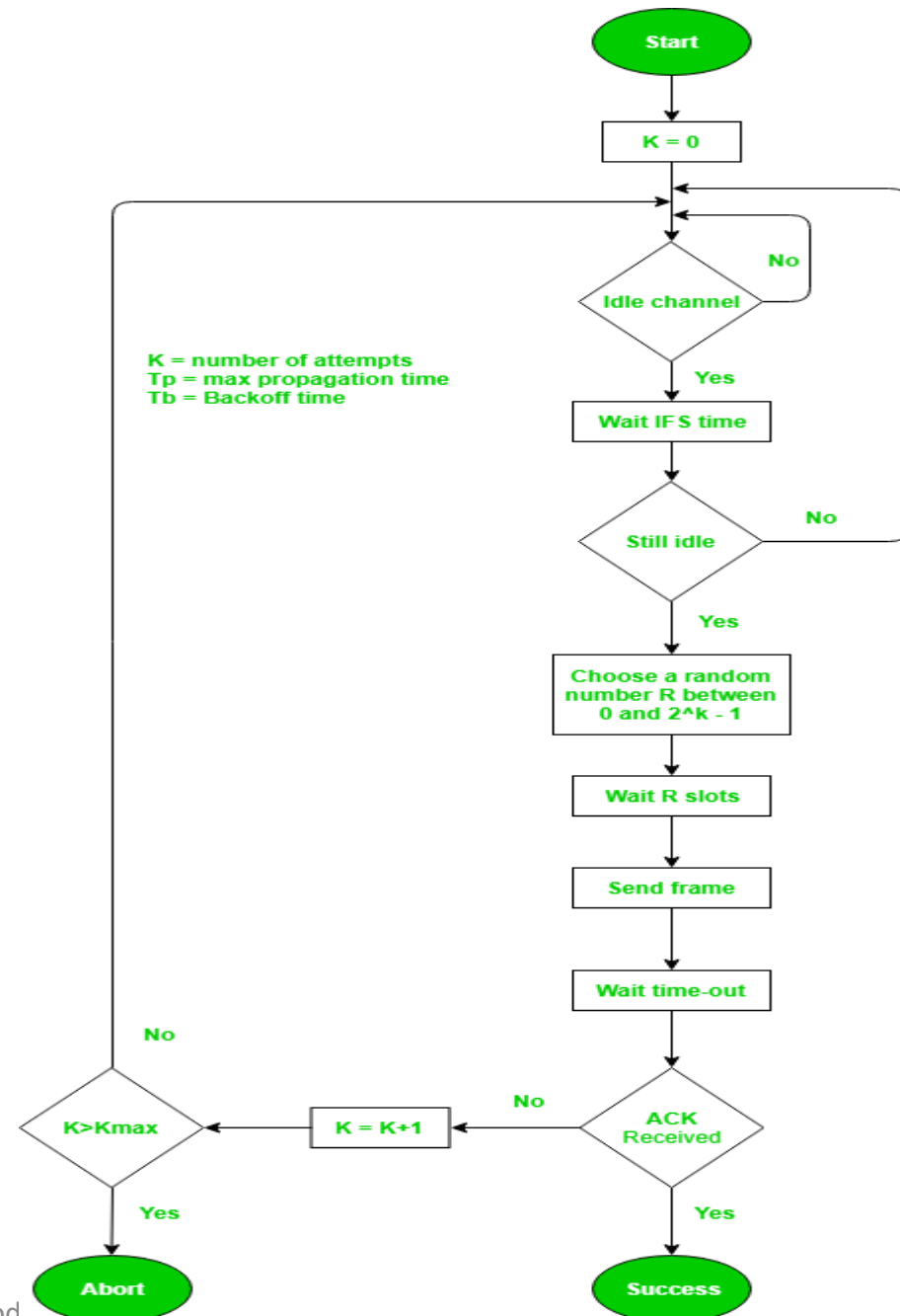
These are three types of avoiding strategies:

InterFrame Space (IFS) – when the channel or the medium is found idle, sender does not directly transmit the data. Instead, it will wait for a fixed amount of time called interframe space. When the previously transmitted signal will reach the previous station, it will then send the next data. During the IFS, the sent data will reach either one of the stations. If the IFS time is over and the channel is still idle, the next data is sent. In this way, there are not collision faced.

Contention Window – It is the amount of time divided into slots. A station which is **ready to send** frames chooses random number of slots as **wait time**.

Acknowledgements – The positive acknowledgements and time-out timer can help guarantee a successful transmission of the frame.

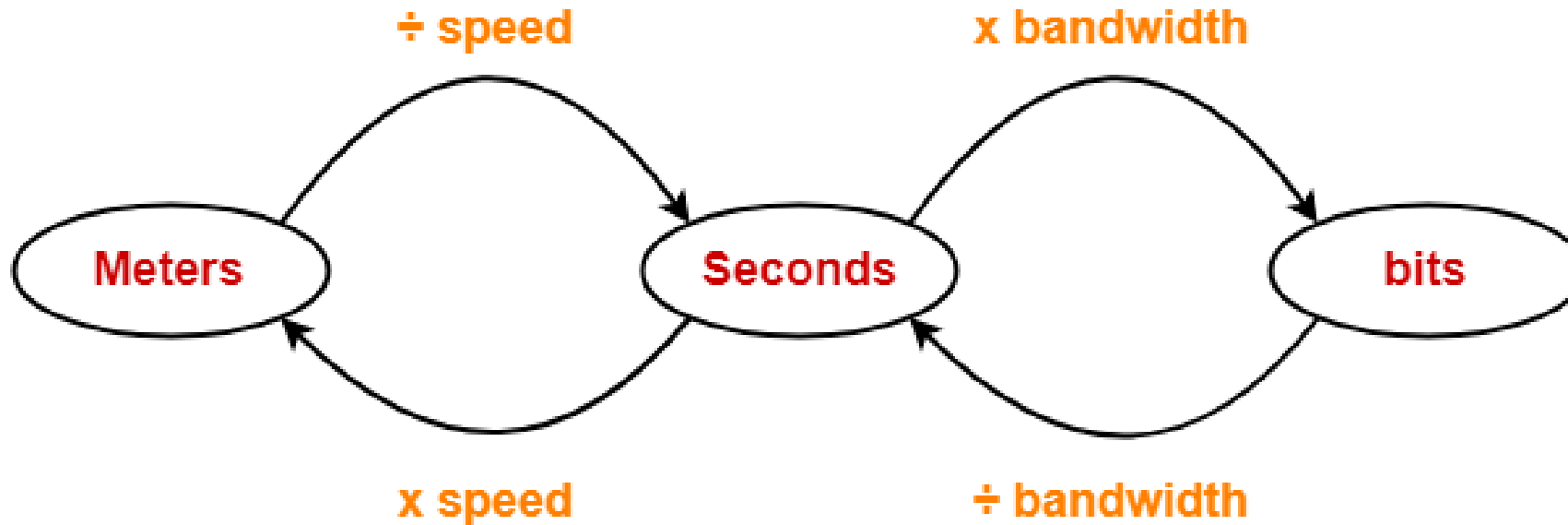
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Difference between CSMA/CD and CSMA/CA

- The CSMA/CD is used in wired local area networks and CSMA/CA used in wireless local area networks and other types of wireless networks.
- The CSMA/CD is standardized in IEEE 802.3 and CSMA/CA is standardized in IEEE 802.11 .
- CSMA/CD will take no action to prevent the transmission collision until it has occurred, while CSMA/CA takes measures to avoid any collision, the latter having no way of knowing whether a collision occurred.
-

*Time Conversions



BW=bits/sec i.e. $B=b/s$

Speed (velocity)=meter/sec i.e. $v=m/s$

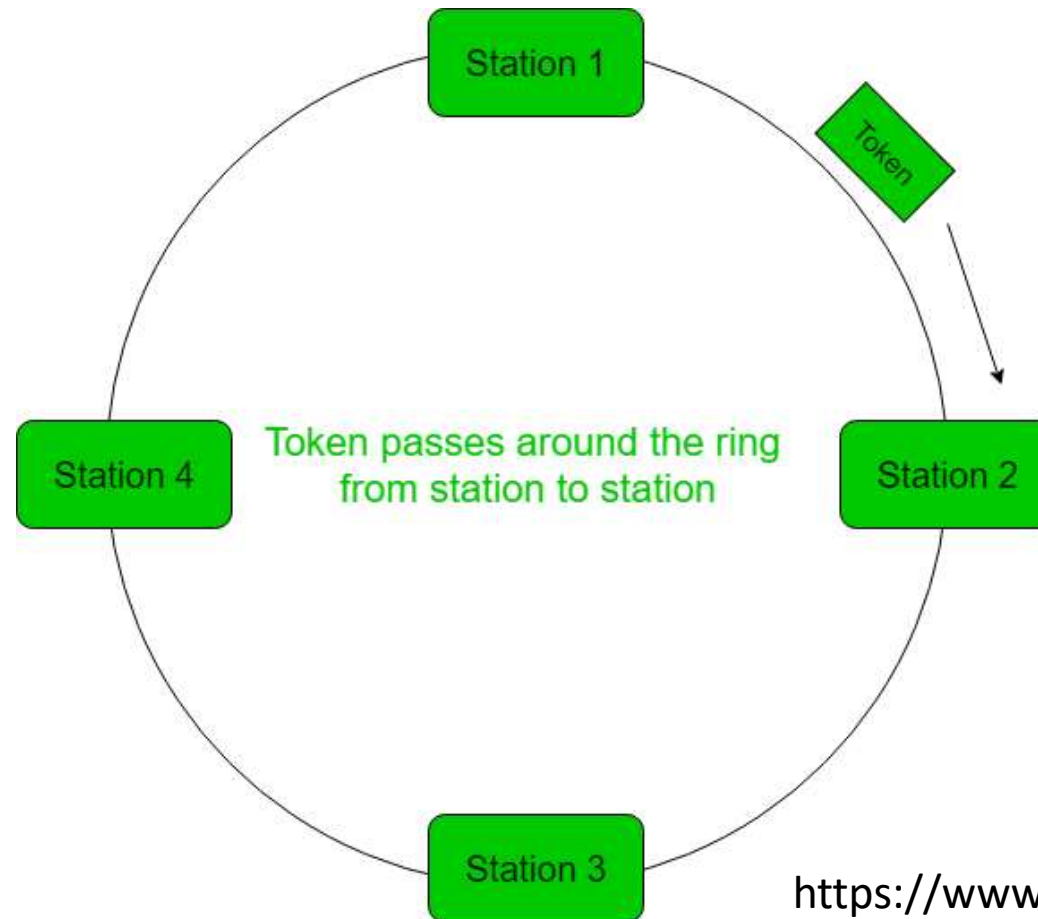
How many times in meters is equivalent to 10bits time if given bandwidth is 4Mbps and velocity is 2×10^8 m/sec.

➤ $(10/B) * V = (10/4 \times 10^6) * 2 \times 10^8 = 500m$

Token Passing

- All the stations are logically connected to each other in the form of a **ring**.
- The access of stations to the transmission link is governed by **a token**.
- A token is a small message/packet composed of a special bit pattern/frame
- A station is allowed to transmit a data packet if and only if it possess the **token** otherwise not.
- If there is **only 1 token**, **only one station** can transmit data packet.
- Each station passes the token to its neighboring station either **clockwise or anti-clockwise (unidirectional)**. Therefore there is no collision.
- There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.

Token Passing...



<https://www.youtube.com/watch?v=0QQgVsv0YvE>

Token Passing

- Assumptions-

- Each station in the ring has the data to send.
- Each station sends exactly one data packet after acquiring the token.

- **Efficiency** (η) = Useful Time / Total Time

- In one cycle, **Useful time** = Sum of transmission delay of N stations since each station sends 1 data packet = $N \times T_t$
 - **Total Time** = **Cycle time** = $T_p + N \times \text{THT}$, where propagation delay or ring latency (T_p) expressed in **seconds**. Also, $T_p = d/v$; d =length of the ring, v =speed of the bit
 - Token Holding Time (THT) depends on the **strategy** implemented. **Delayed** token reinsertion(DTR) and **early** token reinsertion(ETR) time.
 - E.g. For DTR, $\text{THT} = T_t + T_p$, for ETR, $\text{THT} = T_t$
 - Thus,

$$\text{Efficiency } (\eta) = \frac{N \times T_t}{T_p + N \times \text{THT}}$$

3.7 Ethernet (IEEE 802.3 or CSMA/CD)_(ch-1)

- Ethernet was named for IEEE 802.3 standards in a way that
 - physical media could similarly carry data everywhere throughout the network.
- Ethernet refers to a family of local area network (LAN) technologies that share same main features
- The implementation of the network might be different but basic topology, frame type, and network access method remain the same.

Ethernet History

- 10 Mbps Ethernet became an IEEE standard in 1985 – IEEE 802.3
- **high-speed versions:**
 - 100 Mbps - Fast Ethernet (1995)
 - 1000 Mbps - Gigabit Ethernet (1998)
 - 10 Gbps - 10 Gigabit Ethernet (2002)
 - 100 Gbps - 100 Gigabit Ethernet (2007 / 2010)
 - currently used in about 80-90 % of all LANs

Ethernet Frame Types

- Ethernet is an example of a protocol that works at the Media Access Control layer level.
- Ethernet II frame type
 - **De facto standard frame type** used for IP datagram transmissions over Ethernet networks
 - Has protocol identification field (Type field)
- Ethernet frame types that TCP/IP can use
 - **Ethernet II (802.3)**
 - **Ethernet 802.2 Logical Link Control**

Ethernet II frame Structure

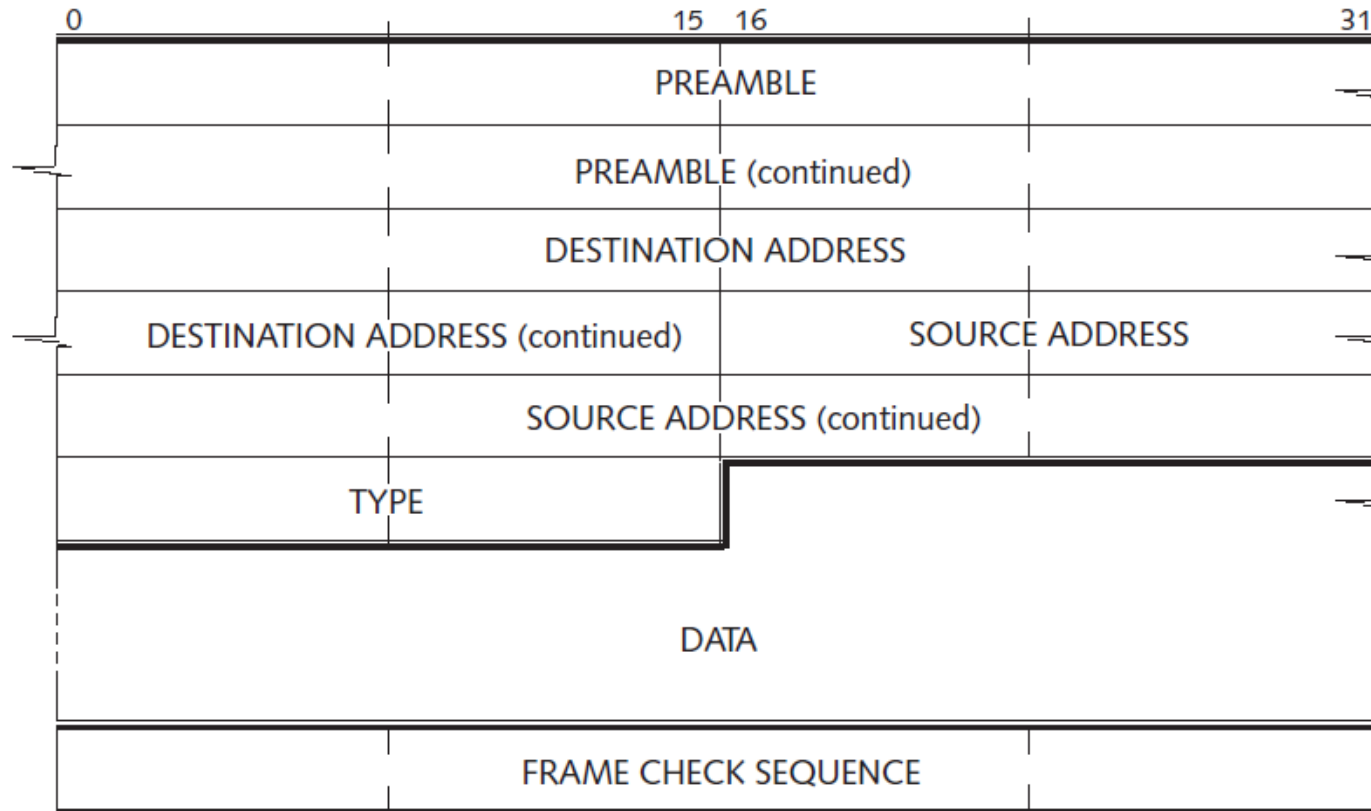


Figure 4-2 Format of an Ethernet II frame

Ethernet II frame Structure...

- Preamble (8 bytes): consists of alternating 1s and 0s. precedes the actual ethernet frame and is not counted as part of the overall frame length. The final byte ends in a pattern....**10101011 SFD**, indicating the start of the destination address.
- Destination Address field(**6** bytes): indicates data link address (hardware address or MAC address) of the destination IP host. DA may be **broadcast, multicast or unicast**
- Source Address field (**6** bytes): indicates sender's hardware address. It can only contain a **unicast** address; can't contain a broadcast or multicast address
- **Type field** (**2** bytes): identifies the protocol that is using this frame type.
- Data field: can be between **46** and **1,500** bytes
- FCS field (**4** bytes): includes the result of the cyclical redundancy check (**CRC**) calculation
- Minimum frame length: **64** bytes and Maximum frame length: **1518** bytes

Type	Protocol
0x0800	IPv4
0x86dd	IPv6
0x0806	Address Resolution Protocol
0x809B	AppleTalk
0x8137	Novell Internetwork Packet Exchange (IPX)

Table 4-1 Assigned protocol types (by number)

IEEE 802.3

IEEE Ethernet standards

Original IEEE	IEEE Shorthand Name	Informal Name(s)	Speed	Typical Cabling
802.3i	10BASE-T	Ethernet	10 Mbps	UTP
802.3u	100BASE-T	Fast Ethernet (Fast E)	100 Mbps	UTP
802.3z	1000BASE-X	Gigabit Ethernet (Gig E, GbE)	1000 Mbps	Fiber
802.3ab	1000BASE-T	Gigabit Ethernet (Gig E, GbE)	1000 Mbps	UTP
802.3ae	10GBASE-X	10 GbE	10 Gbps	Fiber
802.3an	10GBASE-T	10 GbE	10 Gbps	UTP
802.3ba	40GBASE-X	40GbE (40 GigE)	40 Gbps	Fiber
802.3ba	100GBASE-X	100GbE (100 GigE)	100 Gbps	Fiber

Media Standard	Cable Type	Bandwidth Capacity	Maximum Length
10Base2	Coaxial	10 Mbps	185m
10Base5	Coaxial	10 Mbps	500m
10BaseT	UTP (CAT 3 or higher)	10 Mbps	100m
100BaseTX	UTP (CAT 5 or higher)	100 Mbps	100m
10BaseFL	Fibre Optic	10 Mbps	2Km
100BaseFX	Fibre Optic	100 Mbps	HD 400m/FD 2km
1000BaseT	UTP (CAT 6 or higher)	1 Gbps (1000 Mbps)	100m
10GBase-T	UTP (Cat 6a or higher)	10 Gbps (10000 Mbps)	100m
1000BaseSX	Fibre Optic	1 Gbps (1000 Mbps)	MMF 550m
1000BaseLX	Fibre Optic	1 Gbps (1000 Mbps)	MMF 500m/SMF 10km
1000BaseCX	Fibre Optic	1 Gbps (1000 Mbps)	100m
10GbaseSR	Fibre Optic	10 Gbps	300m
10GbaseLR	Fibre Optic	10 Gbps	SMF 10km

Ethernet LLC frame

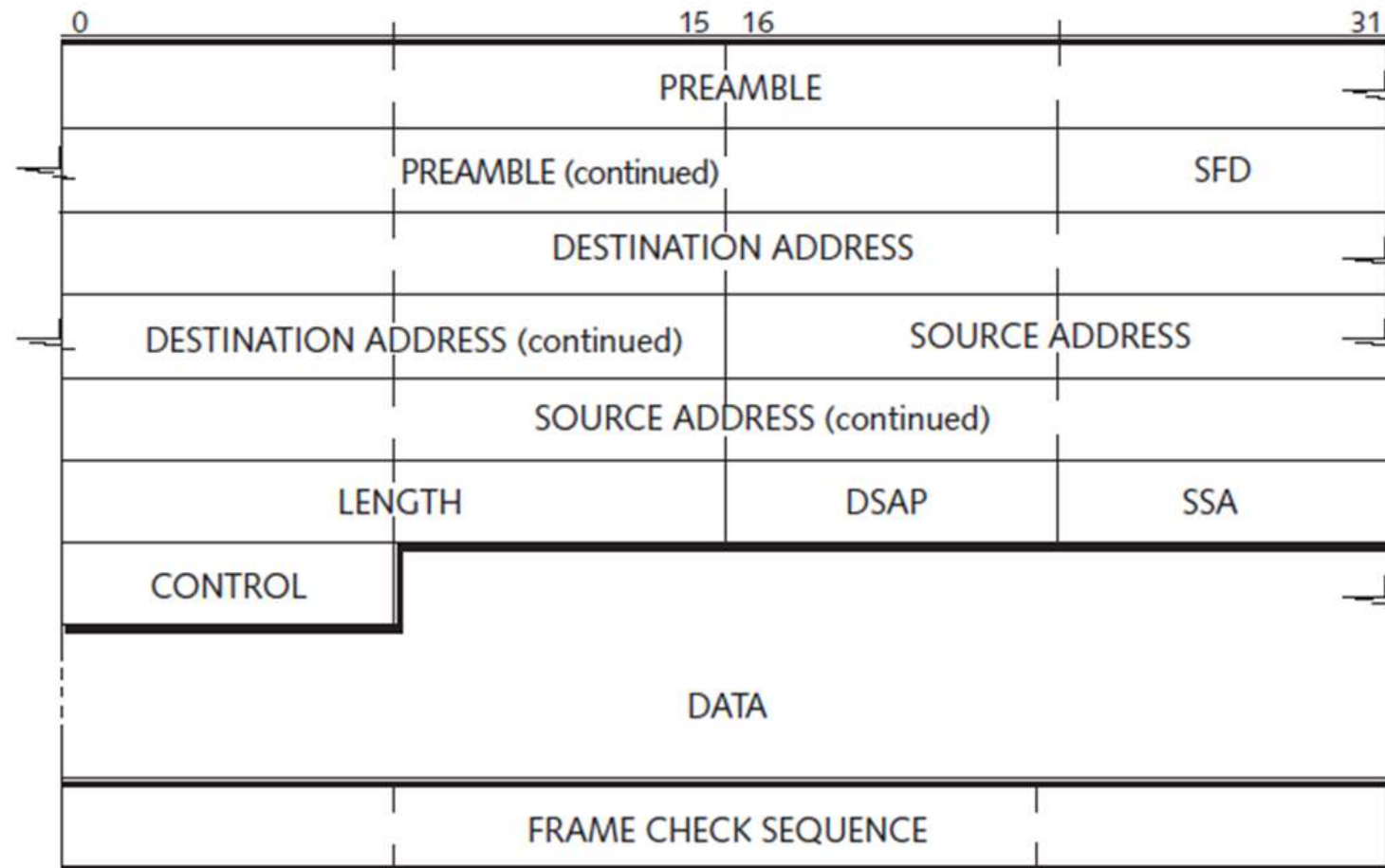


Figure 4-3 Format of an Ethernet 802.2 LLC frame

Ethernet 802.2 LLC Frame Structure

- Preamble: 7 bytes long consists of alternating 1s and 0s
- Start Field Delimiter (SFD) :1 byte used to mark the beginning of the Destination Address field....10101011
- Destination address field: 6 bytes
- Source address field: 6 bytes
- length field: 2 byte; indicates the number of bytes in the data portion of the frame.
- Destination Service access Point (DSAP) field: 1 byte; indicates the destination protocol
- Source service Access Point (SSAP) field: 1 byte; the source protocol
- Control field: 1 byte; ...whether frame is unnumbered format (connectionless) or informational/Supervisory format (for connection oriented and management purposes)
- Data
- Frame Check Sequence (FCS)

- IEEE 802.3(Ethernet),
- 802.4(Token Bus),
- 802.5(Token Ring), and
- 802.11(Wireless LAN).

IEEE 802 Specifications

Table 5-2 IEEE 802 standards

Standard	Name	Explanation
802.1	Internetworking	Covers routing, bridging, and inter-network communications
802.2	Logical Link Control	Covers error control and flow control over data frames
802.3	Ethernet LAN	Covers all forms of Ethernet media and interfaces, from 10 Mbps to 10 Gbps (10 Gigabit Ethernet)
802.4	Token Bus LAN	Covers all forms of token bus media and interfaces
802.5	Token Ring LAN	Covers all forms of token ring media and interfaces
802.6	Metropolitan Area Network	Covers MAN technologies, addressing, and services
802.7	Broadband Technical Advisory Group	Covers broadband networking media, interfaces, and other equipment
802.8	Fiber-Optic Technical Advisory Group	Covers use of fiber-optic media and technologies for various networking types
802.9	Integrated Voice/Data Networks	Covers integration of voice and data traffic over a single network medium

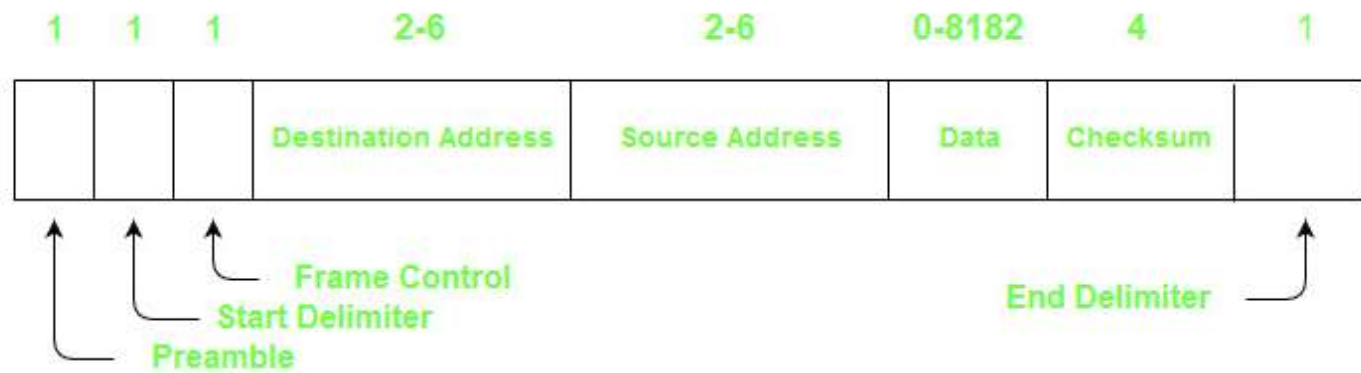
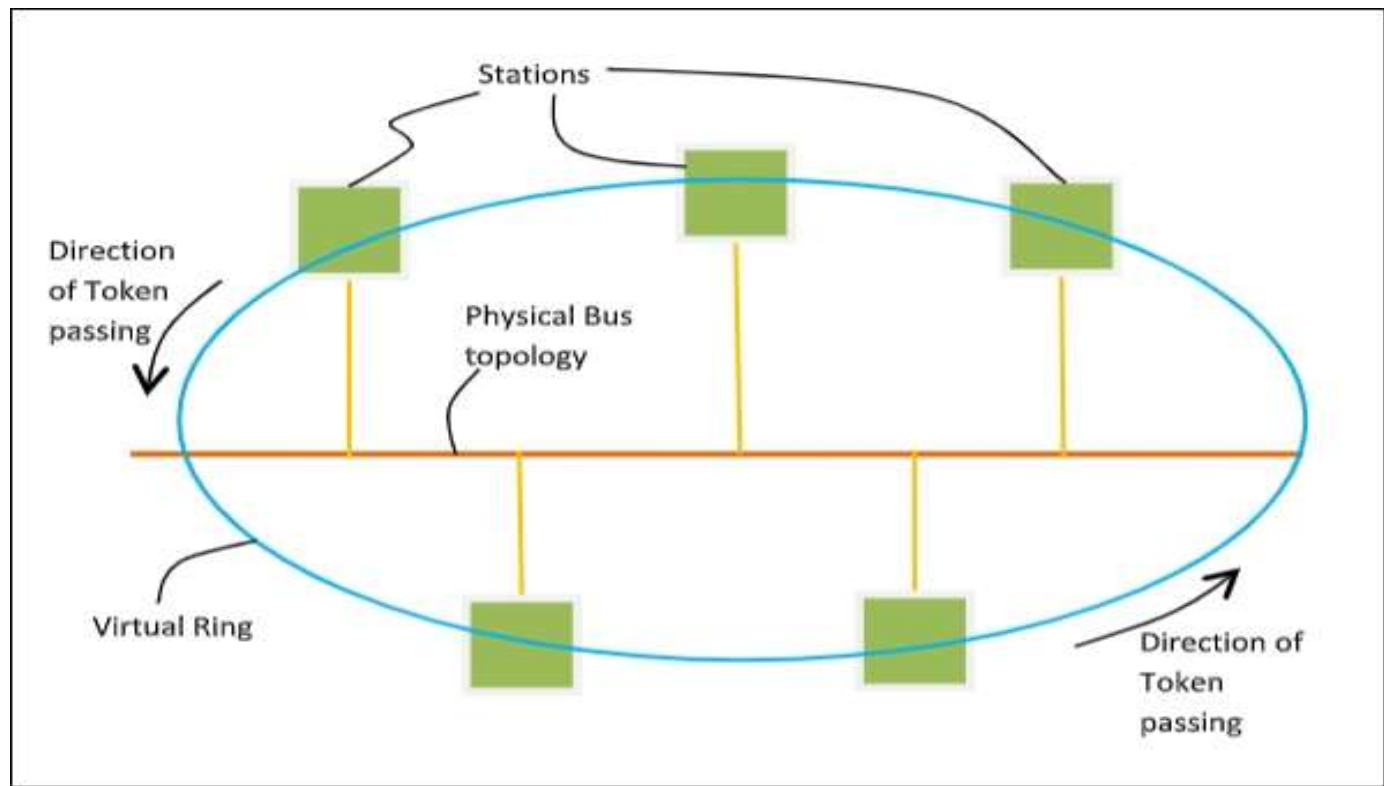
IEEE 802 Specifications (continued)

Table 5-2 IEEE 802 standards (continued)

Standard	Name	Explanation
802.10	Network Security	Covers network access controls, encryption, certification, and other security topics
802.11	Wireless Networks	Sets standards for wireless networking for many different broadcast frequencies and techniques
802.12	High-Speed Networking	Covers a variety of 100 Mbps-plus technologies, including 100VG-AnyLAN
802.13		Unused standard number
802.14	Defunct working group	Specifies data transports over cable TV
802.15	Wireless PAN	Covers the standards for wireless personal area networks
802.16	Wireless MAN	Covers wireless metropolitan area networks
802.17	Resilient Packet Ring	Covers emerging standards for very high-speed, ring-based LANs and MANs
802.18	Wireless Advisory Group	A technical advisory group that monitors radio-based wireless standards
802.19	Coexistence Advisory Group	A group that addresses issues of coexistence with current and developing standards
802.20	Mobile Broadband Wireless	A group working to enable always-on multivendor mobile broadband wireless access

3.6 IEEE 802.4 Token Bus

- **Token Bus (IEEE 802.4)** is a popular standard for the token passing LANs.
- In a token bus LAN, the physical media is a bus or a tree and a **logical ring (Virtual ring)** is created using **coaxial cable**.
- The token is passed from one user to other in a sequence (clockwise or anticlockwise).
- Each station knows the address of the station to its “left” and “right” as per the sequence in the logical ring.
- A station can only transmit data when it has the token. The working of token bus is somewhat similar to **Token Ring (802.5)**.



Frame Format

- **Preamble** – It is used for bit synchronization. It is 1 byte field.
- **Start Delimiter** – These bits marks the beginning of frame. It is 1 byte field.
- **Frame Control** – This field specifies the type of frame – data frame and control frames. It is 1 byte field.
- **Destination Address** – This field contains the destination address. It is 2 to 6 bytes field.
- **Source Address** – This field contains the source address. It is 2 to 6 bytes field.
- **Data** –The Data field carries the actual data and it may be 8182 bytes when 2 byte addresses are used and 8174 bytes for 6 byte addresses.
- **Checksum** – This field contains the checksum bits which is used to detect errors in the transmitted data. It is 4 bytes field.
- **End Delimiter** – This field marks the end of frame. It is 1 byte field.

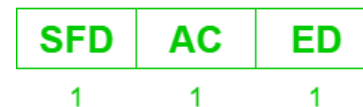
3.6 IEEE 802.5 Token Ring

- The token is passed over a **physical ring** instead of virtual ring.

- **Data Frame**



Token Frame



Topology – Ring topology

Transmission – Unidirectional

Encoding – Differential

Manchester encoding

Access control – Token passing

Data rates – 4 Mbps, 16 Mbps

AC: Access control

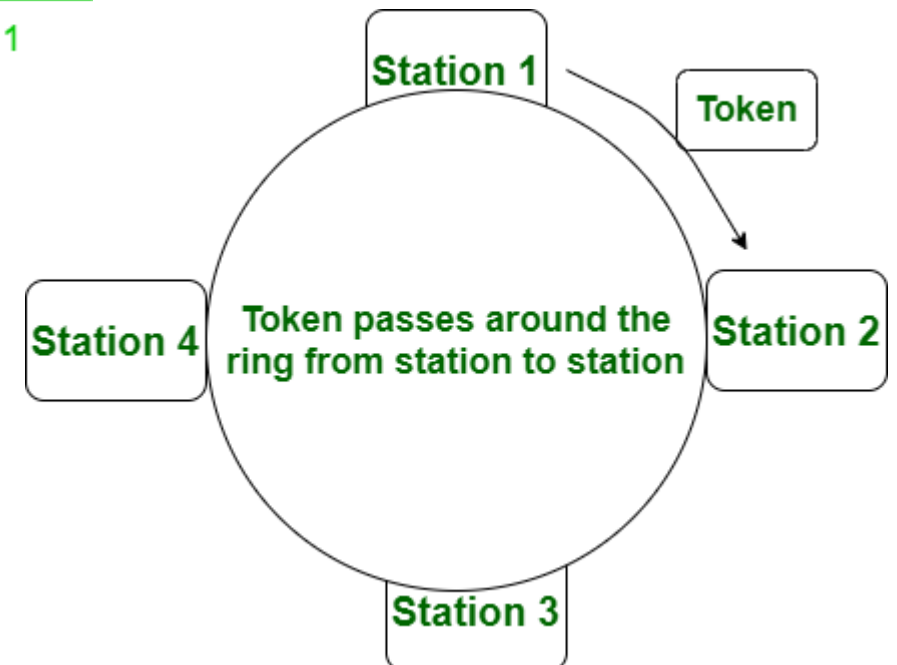
FC: Frame Control

SFD: Start frame delimiter (flag)

EFD: End delimiter (flag)

FS: Frame status

FCS-CRC



Difference between the token bus and the token ring

S.NO	TOKEN BUS NETWORK	TOKEN RING NETWORK
1.	In the token bus network the token is passed along a virtual ring.	While in the token ring network the token is passed over a physical ring.
2.	The token bus network is simply designed for the large factories.	While the token ring network is designed for the offices.
3.	The token bus network is defined by the IEEE 802.4 standard.	While the token ring network is defined by the IEEE 802.5 standard.
4.	Token bus network provides better bandwidth.	While the token ring network does not provide better bandwidth as compared to token bus.
5.	In token bus network, Bus topology is used.	While in token ring network, Star topology is used.
6.	The maximum time it takes to reach the last station in a token bus network cannot be calculated.	While the maximum time to reach the last station in the token ring network can be calculated.

3.8 IEEE 802.11

- **IEEE 802.11** standards are a family of **wireless standards** developed by the **IEEE** for **wireless LAN networks**.
- IEEE 802.11 wireless standards rely on **CSMA/CA access method**.
- Actual operating speeds are less than theoretical speeds.

Standard	Frequency	Theoretical Speed
IEEE 802.11	2.4 GHz	2 Mbps
IEEE 802.11a	5.0 GHz	54 Mbps
IEEE 802.11b	2.4 GHz	11 Mbps
IEEE 802.11g	2.4 GHz	54 Mbps
IEEE 802.11n	2.4 GHz and 5.0 GHz	600 Mbps
802.11ac	5.0 GHz	1300 Mbps

Ethernet (IEEE 802.3) and its evolution, bridged and switched Ethernet

Fiber Distributed Data Interface (FDDI)

- FDDI is an expensive LAN technology that employs a pair of fiber-optic rings. It is a high performance fiber optic token ring.
- FDDI uses fiber-optic cable and is wired in a ring topology
- Uses dual ring: One is primary ring and the second ring /Secondary is used to replace the primary ring in the case of a network failure.
- LAN can be extended up to 200km
- It offers up to 1000 connections.
- FDDI uses token passing as its media-access method and can operate at high speeds(100Mbps).
- Backbone of WAN

FDDI...

- 4B5B line code is used as data encoding technique.
- Frame size is between 17 to 4500 bytes.
- It uses multimode optical fiber with following transmission medium parameters.
Optical source (1300 nm LED), Core diameter (62.5 micron), Cladding diameter (125 micron), numerical aperture (0.275), bandwidth (500 MHz/Km).

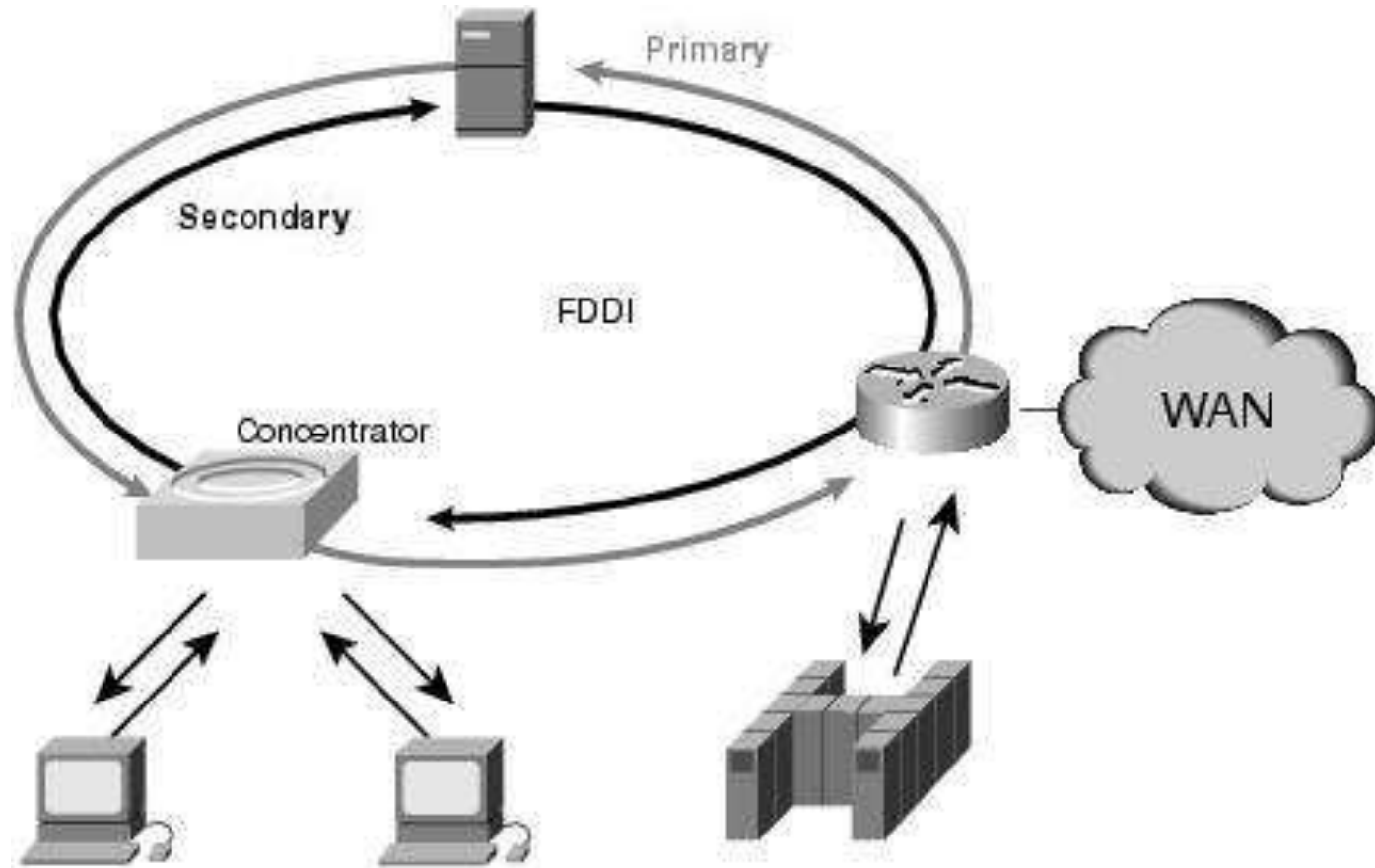
FDDI...

- The FDDI provides high-speed network backbones that can be used to **connect and extend LANs**.
- Like token ring, FDDI also has error-detection and correction capabilities.
- In a normally operating FDDI ring, the token passes by each network device fast.
 - If the token is not seen within the maximum amount of time that it takes to circulate the largest ring, it indicates a network problem.
- **Fiber-optic cable** used with FDDI can support very large volumes of data over large distances.

FDDI...

- FDDI networks are comprised of two physical paths, or "rings," that transfer data in opposite directions.
- The **primary ring** carries **data** between systems, while the **secondary ring** is used for **redundancy**.
- If a system on the network causes an interruption in the primary data path, the secondary ring is used until the primary ring is functional again.
- A variation of FDDI, called **FDDI Full Duplex Technology** (FFDT), uses the secondary ring as an additional primary channel.
- This type of FDDI network has no redundancy, but supports data transfer rates up to 200 Mbps.

FDDI...

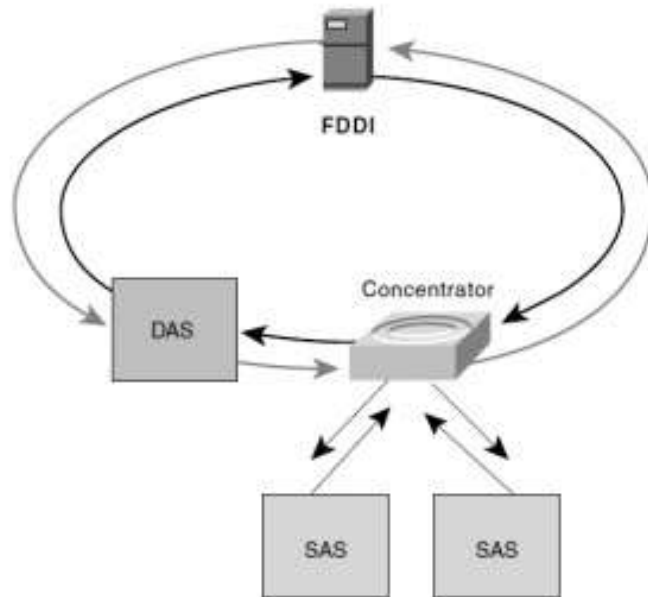


How FDDI Works?

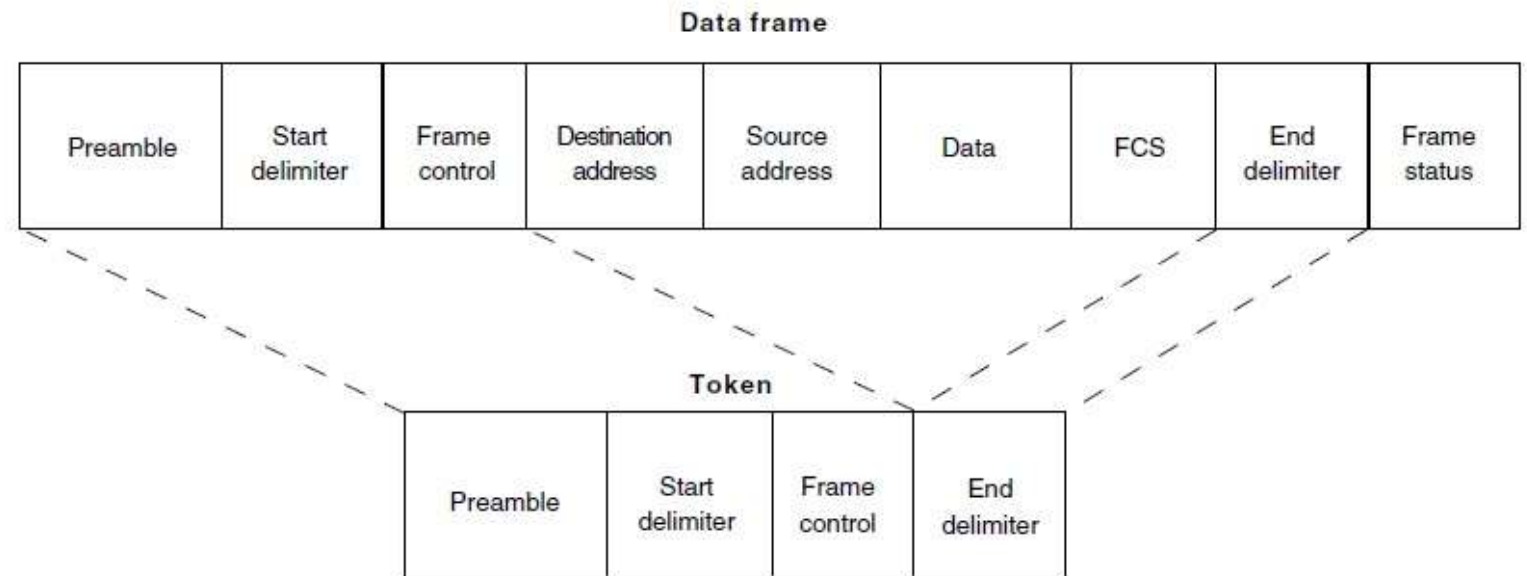
- The primary ring carries data.
- Based on token ring protocol
- The counter-rotating secondary ring can carry data in the opposite direction, but is more commonly reserved as a backup in case the primary ring goes down.
- This provides FDDI with the degree of fault tolerance necessary for network backbones.
- In the event of a failure on the primary ring, FDDI automatically reconfigures itself to use the secondary ring as shown in the illustration.
- Faults can be located and repaired using a fault isolation technique called **beaconing**. However, the secondary ring can also be configured for carrying data, extending the maximum potential bandwidth to 200 Mbps.

Two FDDI stations

- FDDI defines four types of devices: single-attachment station (SAS), dual-attachment station (DAS), single-attached concentrator (SAC), and dual-attached concentrator (DAC).
- **Single-attached stations (SAS)**
- Connect to either the primary or secondary ring using M ports.
- Single-attached stations are used primarily to connect Ethernet LANs or individual servers to FDDI backbones.
- An SAS attaches to only one ring (the primary) through a concentrator. One of the primary advantages of connecting devices with SAS attachments is that the devices will not have any effect on the FDDI ring if they are disconnected or powered off.
- **Dual-attached stations (DAS) :**
- Dual-attached FDDI uses both rings, with the secondary ring serving as a backup for the primary.
- Dual-attached FDDI is used primarily for network backbones that require fault tolerance.
- Devices using DAS connections will affect the rings if they are disconnected or powered off.
- Single-attached stations can be connected to dual-attached FDDI backbones using a dual-attached device called a concentrator or multiplexer.



A Concentrator Attaches to Both the Primary and Secondary Rings



The FDDI Frame Is Similar to That of a Token Ring Frame

Advantages of FDDI

- ➡ It offers higher bandwidth (upto 250 Gbps). Hence it can handle data rates upto 100 Mbps.
- ➡ It offers high security as it is difficult to eavesdrop on fiber optic link.
- ➡ Fiber optic cable does not break as easily as other types of cables.
- ➡ Fiber optic cables are not susceptible to EMI (Electromagnetic Interference).
- ➡ Fiber optic cables transmit signals over greater distances of about 200 Kms.
- ➡ Fiber optic cables weigh less compare to copper wire of similar bandwidth.
- ➡ FDDI uses multiple tokens to improve network speed.
- ➡ It is possible to provide priority to the work stations connected in the chain. Hence based on the requirement some stations are bypassed to provide faster service to the rest.
- ➡ FDDI can isolate faulty nodes with use of wiring concentrators for instantaneous re-routing. Wiring concentrators function as centralized cabling connection devices for workstations.

Disadvantages of FDDI

- FDDI is complex. Hence installation and maintenance require great deal of expertise.
- FDDI is costly. This is because fiber optic cable, adapters and concentrators are very expensive.

3.9 VLAN

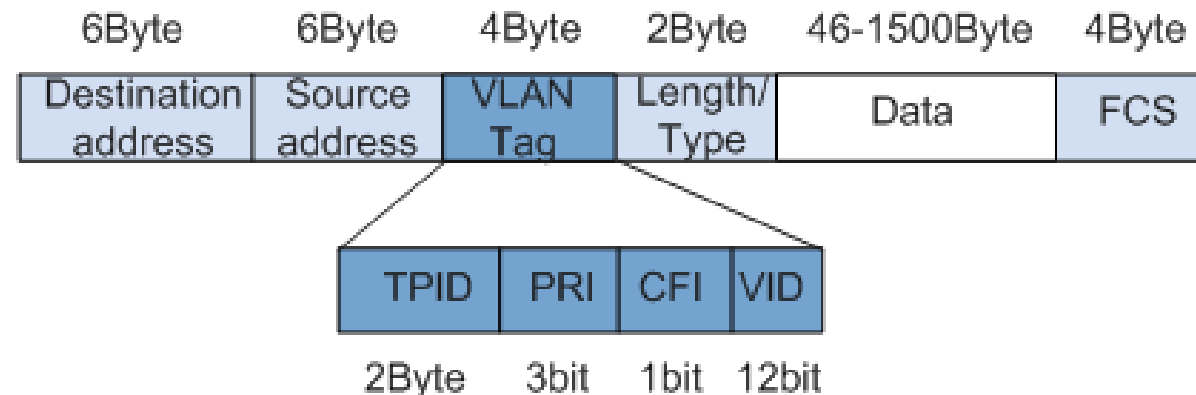
- A VLAN is a logical partition of a layer 2 network (usually via a switch)
- Multiple partitions can be created, allowing for multiple VLANs to coexist
- Each VLAN is a **broadcast domain**, usually with its own IP network
- VLANs are mutually isolated and packets can only pass between them via a router
- The hosts grouped within a VLAN are unaware of the VLAN's existence (only switch knows)

VLAN

- VLANs are used to divide a physical LAN into multiple **broadcast domains** to isolate services with the aim of improving the security and management of the network.
- Hosts within a VLAN can directly communicate only with other hosts in the same VLAN and must use a router to communicate with hosts in other VLANs.
- The development of switching technology also accelerates the application of new switching technologies (VLANs). By dividing the enterprise network into virtual network VLAN segments, network management and network security can be enhanced to control unnecessary data broadcasting.
- A VLAN is a protocol for addressing the **broadcast problems** and security of Ethernet, which adds VLAN headers to Ethernet frames, divides users into smaller workgroups with **VLAN IDs**, and restricts user visits between different workgroups, and each workgroup is a virtual LAN.
- The advantage of a virtual LAN is that it can limit the scope of the broadcast and can form a virtual workgroup to dynamically manage the network.

VLAN frame tagging

- Frame tagging is the process of adding VLAN identification header to the frame
- It is used to properly transmit multiple VLAN frames through a trunk link
- Switches tag frames to identify the VLAN to that they belong. IEEE 802.1Q is a popular tagging example.
- Switches add VLAN tags to the frame before placing them into trunk links and remove the tags before forwarding frames through non-trunk parts.



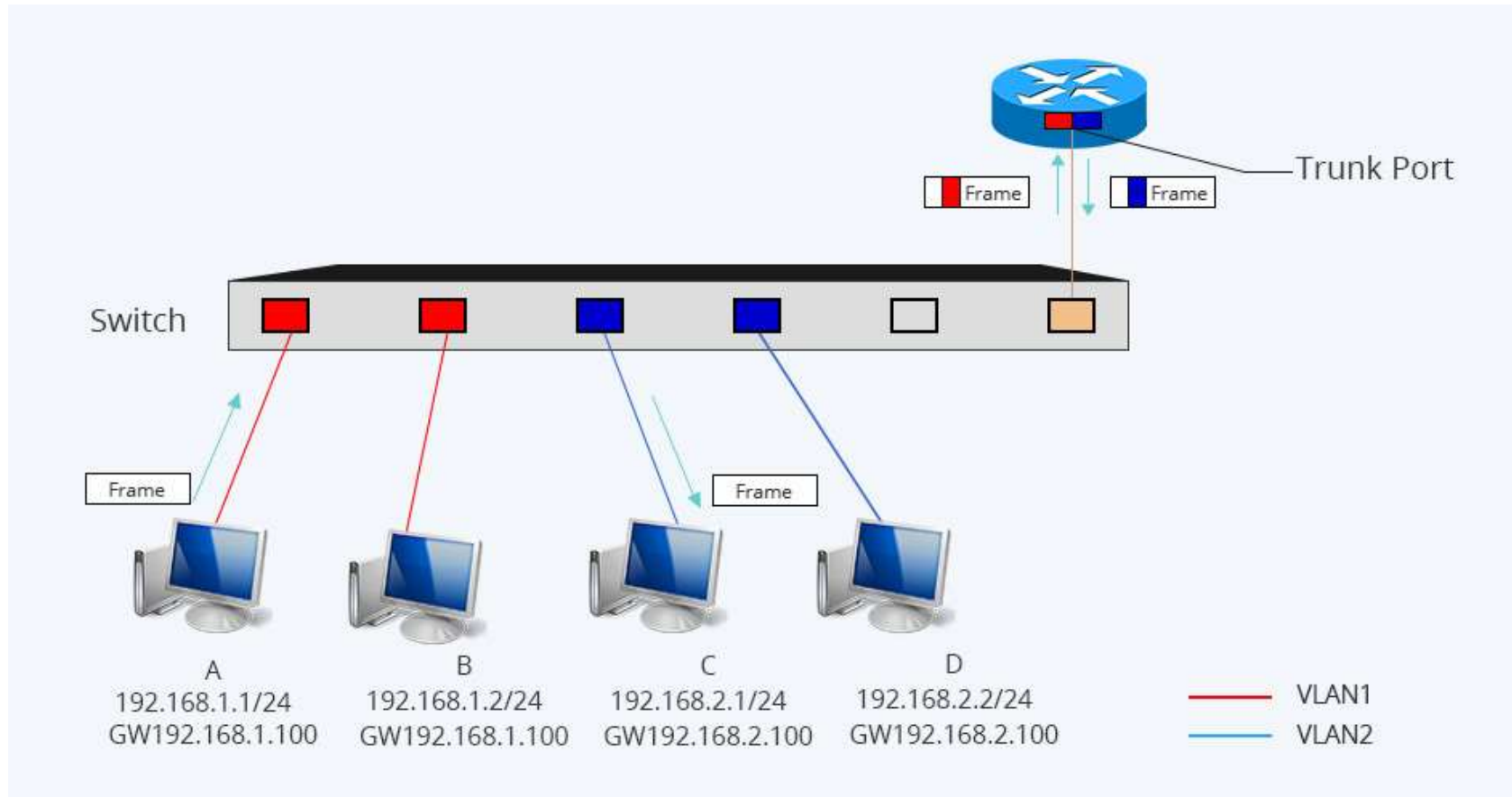
Types of VLANs

- **Static VLAN-** also referred to as **port-based** VLAN,
 - needs a network administrator to assign (manual assignment) the ports on a network switch to a virtual network;
 - Once ports are assigned, there are always associated with their pre-assigned VLANs
 - 1 port only belongs to 1 VLAN
- **Dynamic VLAN (MAC based)-**
 - Dynamic VLAN is defined based on device instead of port location.
 - DVLAN is usually MAC based i.e. VLAN membership is defined according to device's MAC address (Also, We can define based on IP address of device as we want to)
 - Requires a central server called the VLAN member policy server, or simply VMPS server
 - The server contains a database of MAC addresses, along with their associated with their VLANs
 - Changing port location doesn't change a device's VLAN membership
 - 1 port can be a member of multiple VLANs

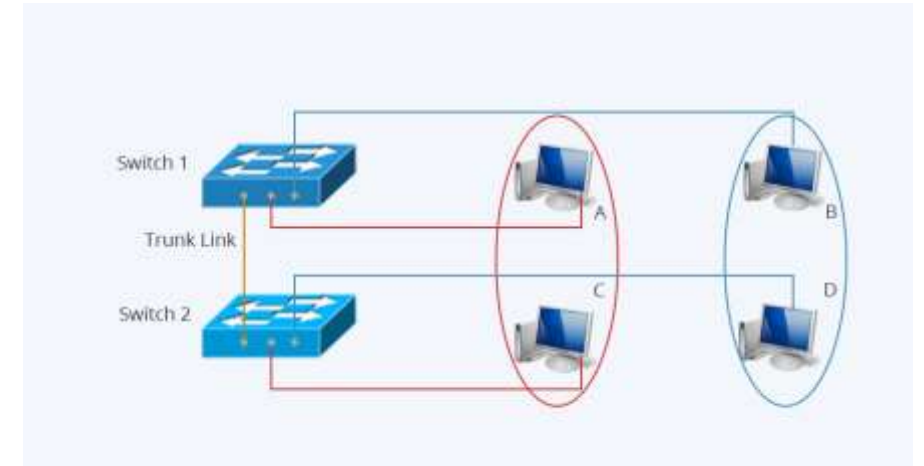
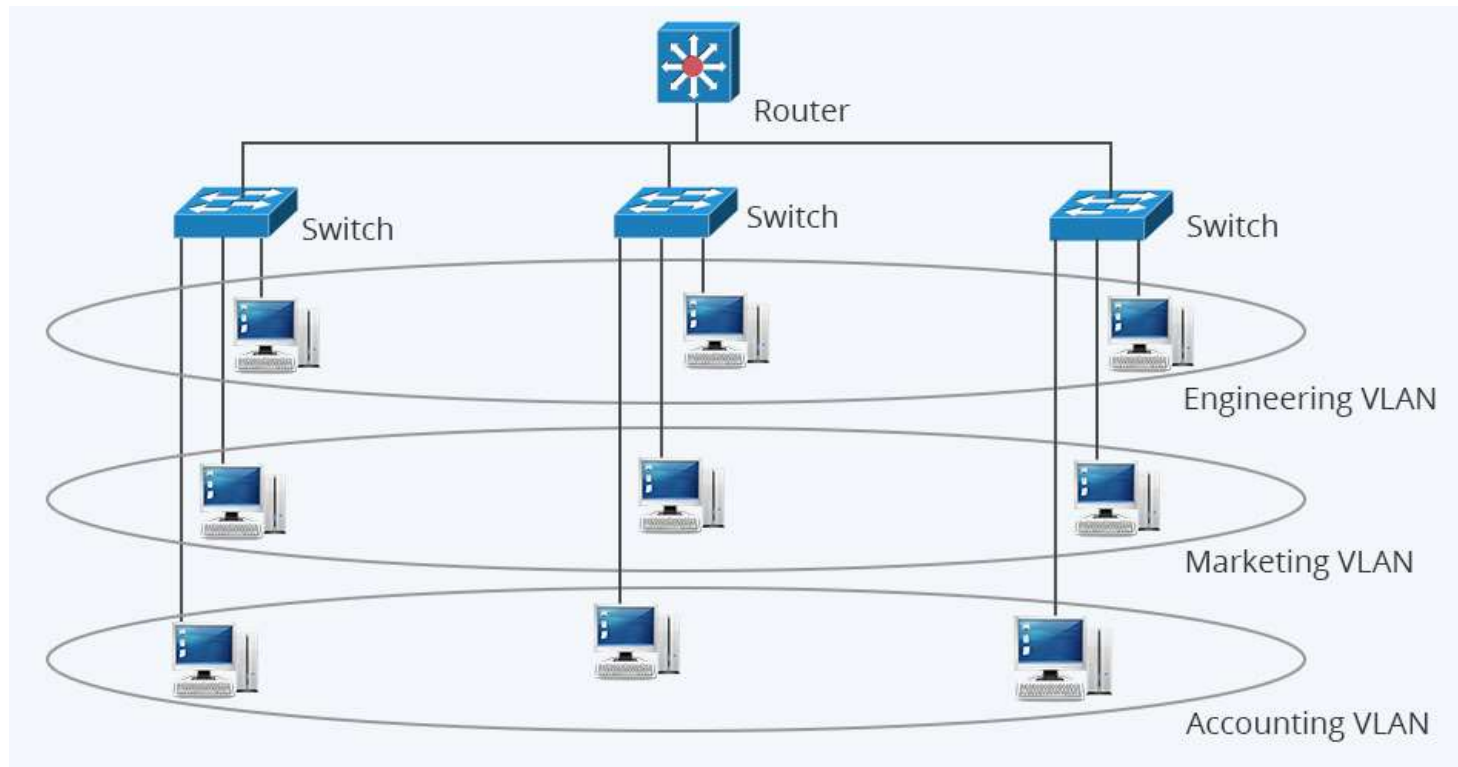
Advantages of dynamic VLAN

- 1. Flexibility: with dynamic VLAN, we can control device on a central server instead of configuring each individual port of all switches
- For a large organization, DVLAN could simplify the network management
- 2. Security: with dynamic VLAN, each device is connected to a switch is checked against the server database for its membership. Therefore no random device could be able to plug in to a switch and get access to the network
- Dynamic VLAN is flexible but more complex and many administrators still refer static VLAN
- <https://www.youtube.com/watch?v=tEM7fus03Ps>

VLAN network diagram



VLAN Segments



VLAN...

- A **broadcast domain** is a network segment which if a/one device broadcast a packet then all the devices in the same broadcast domain will receive it.
- The devices in the same broadcast domain will receive all the broadcast packet but it is limited to **switches** only as **routers don't forward out the broadcast packet**.
- To forward out the packets to different VLAN (from one VLAN to another) or broadcast domain, **inter VLAN routing** is needed.
- Through VLAN, different small size sub networks are created which are comparatively easy to handle.

Why VLAN?(advantages)

- 1. segmentation
- Problems:
 - Traffic congestion and poor performance
 - Security concerns
- Solutions
 - A. We can buy more switches and routers
 - B. We can create two VLANs with one switch(physically the two department share one physical switch but virtually they are two different independent VLANs.
We can also add one router between two switch so that two VLAN can talk each other using a technique call trunking/trunk
 - C. We can use a multilayer switch to create two independent VLANs (No need of Physical router but a virtual router is connected between two multilayer switch –can talk each other using Inter V-LAN routing techniques)
- 2. the simplicity of network design and deployment
- 3. Easier troubleshooting and management

Advantages...

2. The simplicity of network design and deployment/ formation of virtual groups

- With VLANs, Users in the same department can be located indifferent buildings or floors.
- Devices with different functions/missions can be connected to the same switch.
- Without VLANs reorganizing users/computers needs relocating them or rewiring physical data links which would requires hours of labour and huge budget for any organization

3. Easier troubleshooting and management/ Flexibility

VLAN provide flexibility to add, remove the number of host we want

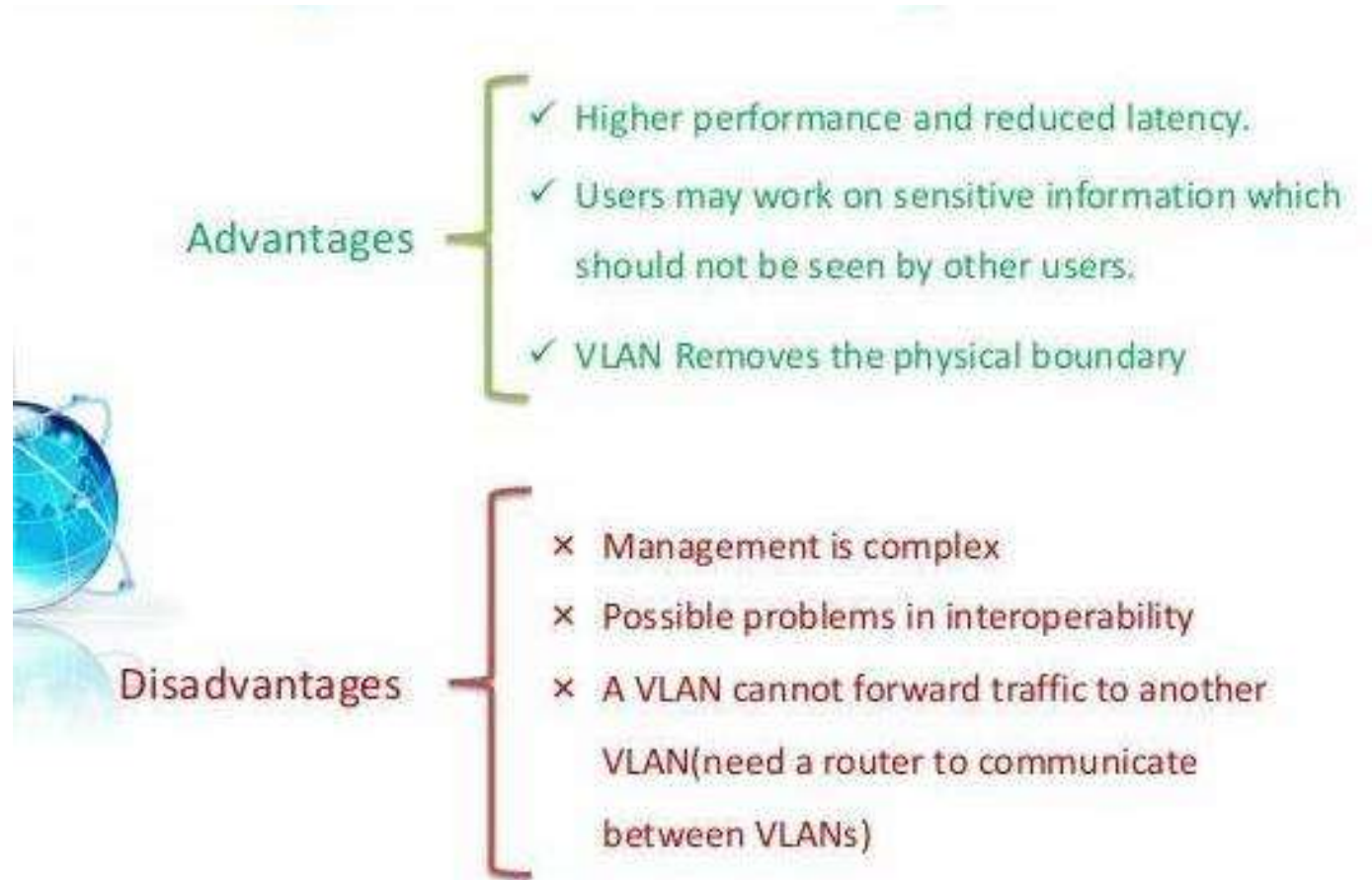
4. Better performance –The network traffic is full of broadcast and multicast. VLAN reduces the need to send such traffic to unnecessary destination. e.g.-If the traffic is intended for 2 users but as 10 devices are present in the same broadcast domain therefore all will receive the traffic i.e. wastage of bandwidth but if we make VLANs, then the broadcast or multicast packet will go to the intended users only.

Advantages...

4. **Security** – In the same network, sensitive data can be broadcast which can be accessed by the outsider but by creating VLAN, we can control broadcast domains, set up firewalls, restrict access. Also, VLANs can be used to inform the network manager of an intrusion. Hence, VLANs greatly enhance network security.
5. **Cost reduction** – VLANs can be used to create broadcast domains which eliminate the need for expensive routers.
6. Shrink broadcast domains
 - By using VLAN, the number of small size broadcast domain can be increased which are easy to handle as compared to a bigger broadcast domain.
7. Improved IT staff efficiency

The **disadvantage** of this method is **routers** usually take more time to process incoming data compared to a bridge or a switch.

Advantages and Disadvantages



3.10 Virtual circuit switching: ATM, MPLS